

# FCOS: Fully Convolutional One-Stage Object Detection

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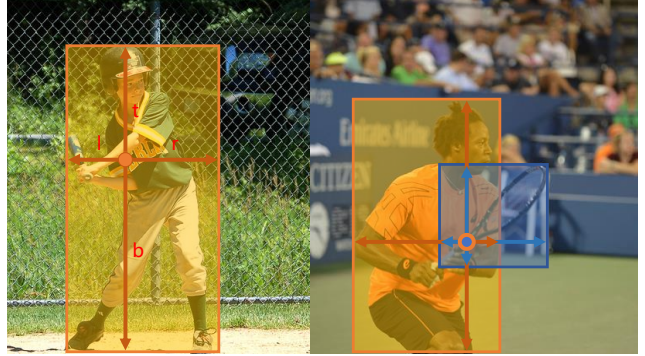
## Abstract

We propose a fully convolutional one-stage object detector (FCOS) to solve object detection in a per-pixel prediction fashion, analogue to semantic segmentation. Almost all state-of-the-art object detectors such as RetinaNet, SSD, YOLOv3, and Faster R-CNN rely on pre-defined anchor boxes. In contrast, our proposed detector FCOS is anchor box free, as well as proposal free. By eliminating the pre-defined set of anchor boxes, FCOS completely avoids the complicated computation related to anchor boxes such as calculating overlapping during training. More importantly, we also avoid all hyper-parameters related to anchor boxes, which are often very sensitive to the final detection performance. With the only post-processing non-maximum suppression (NMS), FCOS with ResNeXt-64x4d-101 achieves 44.7% in AP with single-model and single-scale testing, surpassing previous one-stage detectors with the advantage of being much simpler. For the first time, we demonstrate a much simpler and flexible detection framework achieving improved detection accuracy. We hope that the proposed FCOS framework can serve as a simple and strong alternative for many other instance-level tasks. Code is available at:

[tinyurl.com/FCOSv1](https://tinyurl.com/FCOSv1)

## 1. Introduction

Object detection is a fundamental yet challenging task in computer vision, which requires the algorithm to predict a bounding box with a category label for each instance of interest in an image. All current mainstream detectors such as Faster R-CNN [24], SSD [18] and YOLOv2, v3 [23] rely on a set of pre-defined anchor boxes and it has long been believed that the use of anchor boxes is the key to detectors' success. Despite their great success, it is important to note that anchor-based detectors suffer some drawbacks: 1) As shown in [15, 24], detection performance is sensitive to the sizes, aspect ratios and number of anchor boxes. For example, in RetinaNet [15], varying these hyper-parameters affects the performance up to 4% in AP on the COCO bench-



**Figure 1** – As shown in the left image, FCOS works by predicting a 4D vector  $(l, t, r, b)$  encoding the location of a bounding box at each foreground pixel (supervised by ground-truth bounding box information during training). The right plot shows that when a location residing in multiple bounding boxes, it can be ambiguous in terms of which bounding box this location should regress.

mark [16]. As a result, these hyper-parameters need to be carefully tuned in anchor-based detectors. 2) Even with careful design, because the scales and aspect ratios of anchor boxes are kept fixed, detectors encounter difficulties to deal with object candidates with large shape variations, particularly for small objects. The pre-defined anchor boxes also hamper the generalization ability of detectors, as they need to be re-designed on new detection tasks with different object sizes or aspect ratios. 3) In order to achieve a high recall rate, an anchor-based detector is required to densely place anchor boxes on the input image (e.g., more than 180K anchor boxes in feature pyramid networks (FPN) [14] for an image with its shorter side being 800). Most of these anchor boxes are labelled as negative samples during training. The excessive number of negative samples aggravates the imbalance between positive and negative samples in training. 4) Anchor boxes also involve complicated computation such as calculating the intersection-over-union (IoU) scores with ground-truth bounding boxes.

Recently, fully convolutional networks (FCNs) [20] have achieved tremendous success in dense prediction tasks such as semantic segmentation [20, 28, 9, 19], depth estimation

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# FCOS: 全卷积单阶段目标检测

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## 摘要

We propose a fully convolutional one-stage object detector (FCOS) to solve object detection in a per-pixel prediction fashion, analogue to semantic segmentation. Almost all state-of-the-art object detectors such as RetinaNet, SSD, YOLOv3, and Faster R-CNN rely on pre-defined anchor boxes. In contrast, our proposed detector FCOS is anchor box free, as well as proposal free. By eliminating the pre-defined set of anchor boxes, FCOS completely avoids the complicated computation related to anchor boxes such as calculating overlapping during training. More importantly, we also avoid all hyper-parameters related to anchor boxes, which are often very sensitive to the final detection performance. With the only post-processing non-maximum suppression (NMS), FCOS with ResNeXt-64x4d-101 achieves 44.7% in AP with single-model and single-scale testing, surpassing previous one-stage detectors with the advantage of being much simpler. For the first time, we demonstrate a much simpler and flexible detection framework achieving improved detection accuracy. We hope that the proposed FCOS framework can serve as a simple and strong alternative for many other instance-level tasks. Code is available at:

[tinyurl.com/FCOSv1](https://github.com/whuang33/FCOS)

## 1. 引言

目标检测是计算机视觉中一项基础且具有挑战性的任务，它要求算法为图像中每个感兴趣的实例预测一个带有类别标签的边界框。当前所有主流检测器，如Faster R-CNN [24]、SSD [18] 和 YOLOv2、v3 [23]，都依赖于一组预定义的锚框和it has long been believed that the use of anchor boxes is the key to detectors' success。尽管它们取得了巨大成功，但值得注意的是，基于锚框的检测器存在一些缺点：1) 如[15, 24]所示，检测性能对锚框的尺寸、宽高比和数量非常敏感。例如，在RetinaNet [15]中，调整这些超参数会使COCO基准上的AP性能变化高达4%。

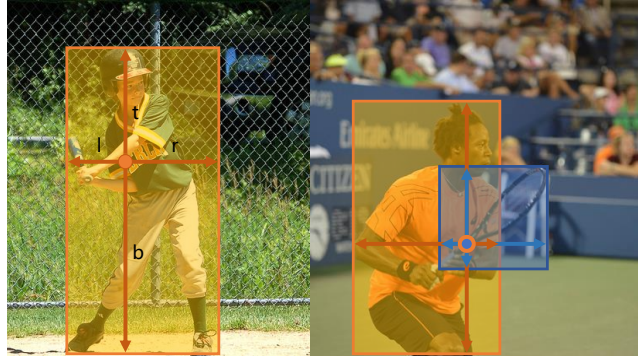


图1 – 如左图所示，FCOS的工作原理是在每个前景像素点预测一个4D向量 ( $\{v^*\}$ )，该向量编码了边界框的位置（训练时通过真实边界框信息进行监督）。右图表明，当某个位置位于多个边界框内时，该位置应回归到哪个边界框可能会产生歧义。

标记[16]。因此，在基于锚点的检测器中，这些超参数需要仔细调整。2) 即使经过精心设计，由于锚框的尺度和长宽比是固定的，检测器在处理形状变化较大的候选物体时仍会遇到困难，特别是对于小物体。预定义的锚框还限制了检测器的泛化能力，因为需要在具有不同物体尺寸或长宽比的新检测任务上重新设计它们。3) 为了实现高召回率，基于锚点的检测器需要在输入图像上密集放置锚框 (e.g., 对于短边为800的图像，特征金字塔网络 (FPN) [14]中超过18万个锚框)。在训练过程中，这些锚框大多被标记为负样本。过多的负样本加剧了训练中正负样本之间的不平衡。4) 锚框还涉及复杂的计算，例如计算与真实边界框的交并比 (IoU) 分数。

最近，全卷积网络 (FCNs) [20]在密集预测任务中取得了巨大成功，例如语义分割[20, 28, 9, 19]、深度估计

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[17, 31], keypoint detection [3] and counting [2]. As one of high-level vision tasks, object detection might be the only one deviating from the neat fully convolutional per-pixel prediction framework mainly due to the use of anchor boxes. It is nature to ask a question: *Can we solve object detection in the neat per-pixel prediction fashion, analogue to FCN for semantic segmentation, for example?* Thus those fundamental vision tasks can be unified in (almost) one single framework. We show that the answer is affirmative. Moreover, we demonstrate that, for the first time, the much simpler FCN-based detector achieves even better performance than its anchor-based counterparts.

In the literature, some works attempted to leverage the FCNs-based framework for object detection such as DenseBox [12]. Specifically, these FCN-based frameworks directly predict a 4D vector plus a class category at each spatial location on a level of feature maps. As shown in Fig. 1 (left), the 4D vector depicts the relative offsets from the four sides of a bounding box to the location. These frameworks are similar to the FCNs for semantic segmentation, except that each location is required to regress a 4D continuous vector. However, to handle the bounding boxes with different sizes, DenseBox [12] crops and resizes training images to a fixed scale. Thus DenseBox has to perform detection on image pyramids, which is against FCN’s philosophy of computing all convolutions once. Besides, more significantly, these methods are mainly used in special domain objection detection such as scene text detection [33, 10] or face detection [32, 12], since it is believed that these methods do not work well when applied to generic object detection with highly overlapped bounding boxes. As shown in Fig. 1 (right), the highly overlapped bounding boxes result in an intractable ambiguity: it is not clear w.r.t. which bounding box to regress for the pixels in the overlapped regions.

In the sequel, we take a closer look at the issue and show that with FPN this ambiguity can be largely eliminated. As a result, our method can already obtain comparable detection accuracy with those traditional anchor based detectors. Furthermore, we observe that our method may produce a number of low-quality predicted bounding boxes at the locations that are far from the center of an target object. In order to suppress these low-quality detections, we introduce a novel “center-ness” branch (only one layer) to predict the deviation of a pixel to the center of its corresponding bounding box, as defined in Eq. (3). This score is then used to down-weight low-quality detected bounding boxes and merge the detection results in NMS. The simple yet effective center-ness branch allows the FCN-based detector to outperform anchor-based counterparts under exactly the same training and testing settings.

This new detection framework enjoys the following advantages.

- Detection is now unified with many other FCN-solvable tasks such as semantic segmentation, making it easier to re-use ideas from those tasks.
- Detection becomes proposal free and anchor free, which significantly reduces the number of design parameters. The design parameters typically need heuristic tuning and many tricks are involved in order to achieve good performance. Therefore, our new detection framework makes the detector, particularly its training, *considerably* simpler.
- By eliminating the anchor boxes, our new detector completely avoids the complicated computation related to anchor boxes such as the IOU computation and matching between the anchor boxes and ground-truth boxes during training, resulting in faster training and testing as well as less training memory footprint than its anchor-based counterpart.
- Without bells and whistles, we achieve state-of-the-art results among one-stage detectors. We also show that the proposed FCOS can be used as a Region Proposal Networks (RPNs) in two-stage detectors and can achieve significantly better performance than its anchor-based RPN counterparts. Given the even better performance of the much simpler anchor-free detector, *we encourage the community to rethink the necessity of anchor boxes in object detection*, which are currently considered as the *de facto* standard for detection.
- The proposed detector can be immediately extended to solve other vision tasks with minimal modification, including instance segmentation and key-point detection. We believe that this new method can be the new baseline for many instance-wise prediction problems.

## 2. Related Work

**Anchor-based Detectors.** Anchor-based detectors inherit the ideas from traditional sliding-window and proposal based detectors such as Fast R-CNN [6]. In anchor-based detectors, the anchor boxes can be viewed as pre-defined sliding windows or proposals, which are classified as positive or negative patches, with an extra offsets regression to refine the prediction of bounding box locations. Therefore, the anchor boxes in these detectors may be viewed as *training samples*. Unlike previous detectors like Fast RCNN, which compute image features for each sliding window/proposal repeatedly, anchor boxes make use of the feature maps of CNNs and avoid repeated feature computation, speeding up detection process dramatically. The design of anchor boxes are popularized by Faster R-CNN in its RPNs [24], SSD [18] and YOLOv2 [22], and has become the convention in a modern detector.

However, as described above, anchor boxes result in excessively many hyper-parameters, which typically need

[17, 31]、关键点检测[3]和计数[2]。作为高级视觉任务之一，目标检测可能是唯一偏离了简洁的全卷积逐像素预测框架的任务，这主要归因于锚框的使用。很自然地会提出一个问题：*Can we solve object detection in the neat per-pixel prediction fashion, analogue to FCN for semantic segmentation, for example?* 因此，这些基础的视觉任务可以（几乎）统一在单一的框架中。我们证明答案是肯定的。此外，我们首次证明，基于FCN的更简单的检测器甚至能取得比基于锚框的检测器更好的性能。

在文献中，一些研究尝试利用基于全卷积网络（FCN）的框架进行目标检测，例如DenseBox [12]。具体而言，这些基于FCN的框架直接在特征图某一层级的每个空间位置上预测一个4维向量及一个类别标签。如图1（左）所示，该4维向量描述了边界框四条边相对于该位置的偏移量。这些框架与用于语义分割的FCN相似，区别在于每个位置需要回归一个4维连续向量。然而，为处理不同尺寸的边界框，DenseBox [12]需将训练图像裁剪并缩放到固定尺度。因此DenseBox必须在图像金字塔上进行检测，这违背了FCN一次性计算所有卷积的核心理念。此外更重要的是，这些方法主要应用于特定领域的检测任务，如场景文本检测[33, 10]或人脸检测[32, 12]，因为普遍认为当应用于具有高度重叠边界框的通用目标检测时，这些方法效果不佳。如图1（右）所示，高度重叠的边界框会导致难以处理的歧义：对于重叠区域内的像素，无法明确应针对哪个边界框进行回归。

在后续部分，我们将更深入地探讨这一问题，并证明通过FPN可以很大程度上消除这种模糊性。因此，我们的方法已经能够获得与传统基于锚点的检测器相当的检测精度。此外，我们观察到，在远离目标物体中心的位置，我们的方法可能会产生一些低质量的预测边界框。为了抑制这些低质量的检测结果，我们引入了一个新颖的“中心度”分支（仅包含一层），用于预测像素点与其对应边界框中心的偏差，如公式(3)所定义。该得分随后被用于降低低质量检测边界框的权重，并在非极大值抑制（NMS）中合并检测结果。这个简单而有效的中心度分支使得基于FCN的检测器在完全相同的训练和测试设置下，能够超越基于锚点的检测器。

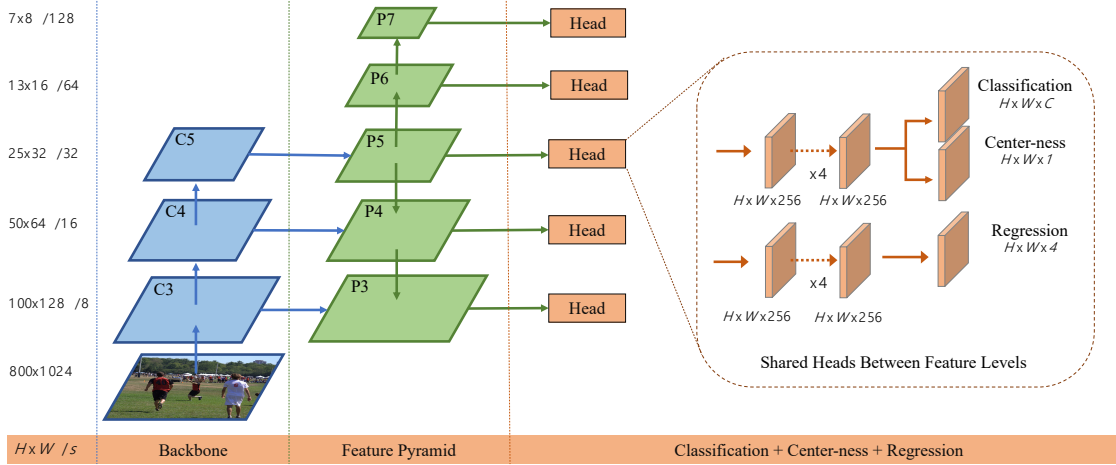
这一新的检测框架具有以下优势。

- 检测现在与许多其他FCN可解决的任务（如语义分割）统一起来，使得从这些任务中复用想法变得更加容易。
- 检测变得无需提议和锚点，这显著减少了设计参数的数量。设计参数通常需要启发式调整，并且涉及许多技巧才能实现良好性能。因此，我们的新检测框架使检测器（尤其是其训练过程）*considerably*更加简单。
- 通过消除锚框，我们的新检测器完全避免了与锚框相关的复杂计算，例如训练期间锚框与真实框之间的IOU计算和匹配，从而比基于锚框的对应方法训练和测试速度更快，训练内存占用也更少。
- 在不添加任何花哨功能的情况下，我们在单阶段检测器中取得了最先进的结果。我们还表明，所提出的FCOS可以用作两阶段检测器中的区域提议网络（RPNs），并且能够比基于锚点的RPN对应物实现显著更好的性能。考虑到更简单的无锚点检测器 *we encourage the community to rethink the necessity of anchor boxes in object detection*（目前被视为检测领域的*de facto*标准）具有更优越的性能。
- 所提出的检测器可以立即扩展到解决其他视觉任务，只需最小修改，包括实例分割和关键点检测。我们相信这种新方法可以成为许多实例级预测问题的新基准。

## 2. 相关工作

基于锚点的检测器。基于锚点的检测器继承了传统滑动窗口和基于提议的检测器（如Fast R-CNN [6]）的思想。在这些检测器中，锚框可视为预定义的滑动窗口或提议区域，它们被分类为正样本或负样本，并通过额外的偏移量回归来细化边界框位置的预测。因此，这些检测器中的锚框可视为*training samples*。与Fast R-CNN等早期检测器需为每个滑动窗口/提议重复计算图像特征不同，锚框利用CNN的特征图，避免了重复的特征计算，从而显著加速了检测过程。锚框的设计由Faster R-CNN在其RPN[24]、SSD[18]和YOLOv2[22]中推广，已成为现代检测器的常规方法。

然而，如上所述，锚框会导致超参数过多，而这些参数通常需要



**Figure 2** – The network architecture of FCOS, where C3, C4, and C5 denote the feature maps of the backbone network and P3 to P7 are the feature levels used for the final prediction.  $H \times W$  is the height and width of feature maps. ‘/s’ ( $s = 8, 16, \dots, 128$ ) is the down-sampling ratio of the feature maps at the level to the input image. As an example, all the numbers are computed with an  $800 \times 1024$  input.

to be carefully tuned in order to achieve good performance. Besides the above hyper-parameters describing anchor shapes, the anchor-based detectors also need other hyper-parameters to label each anchor box as a positive, ignored or negative sample. In previous works, they often employ intersection over union (IOU) between anchor boxes and ground-truth boxes to determine the label of an anchor box (e.g., a positive anchor if its IOU is in  $[0.5, 1]$ ). These hyper-parameters have shown a great impact on the final accuracy, and require heuristic tuning. Meanwhile, these hyper-parameters are specific to detection tasks, making detection tasks deviate from a neat fully convolutional network architectures used in other dense prediction tasks such as semantic segmentation.

**Anchor-free Detectors.** The most popular anchor-free detector might be YOLOv1 [21]. Instead of using anchor boxes, YOLOv1 predicts bounding boxes at points near the center of objects. Only the points near the center are used since they are considered to be able to produce higher-quality detection. However, since only points near the center are used to predict bounding boxes, YOLOv1 suffers from low recall as mentioned in YOLOv2 [22]. As a result, YOLOv2 [22] employs anchor boxes as well. Compared to YOLOv1, FCOS takes advantages of all points in a ground truth bounding box to predict the bounding boxes and the low-quality detected bounding boxes are suppressed by the proposed “center-ness” branch. As a result, FCOS is able to provide comparable recall with anchor-based detectors as shown in our experiments.

CornerNet [13] is a recently proposed one-stage anchor-

free detector, which detects a pair of corners of a bounding box and groups them to form the final detected bounding box. CornerNet requires much more complicated post-processing to group the pairs of corners belonging to the same instance. An extra distance metric is learned for the purpose of grouping.

Another family of anchor-free detectors such as [32] are based on DenseBox [12]. The family of detectors have been considered unsuitable for generic object detection due to difficulty in handling overlapping bounding boxes and the recall being relatively low. In this work, we show that both problems can be largely alleviated with multi-level FPN prediction. Moreover, we also show together with our proposed center-ness branch, the much simpler detector can achieve even better detection performance than its anchor-based counterparts.

### 3. Our Approach

In this section, we first reformulate object detection in a per-pixel prediction fashion. Next, we show that how we make use of multi-level prediction to improve the recall and resolve the ambiguity resulted from overlapped bounding boxes. Finally, we present our proposed “center-ness” branch, which helps suppress the low-quality detected bounding boxes and improves the overall performance by a large margin.

#### 3.1. Fully Convolutional One-Stage Object Detector

Let  $F_i \in \mathbb{R}^{H \times W \times C}$  be the feature maps at layer  $i$  of a backbone CNN and  $s$  be the total stride until the layer. The ground-truth bounding boxes for an input image are



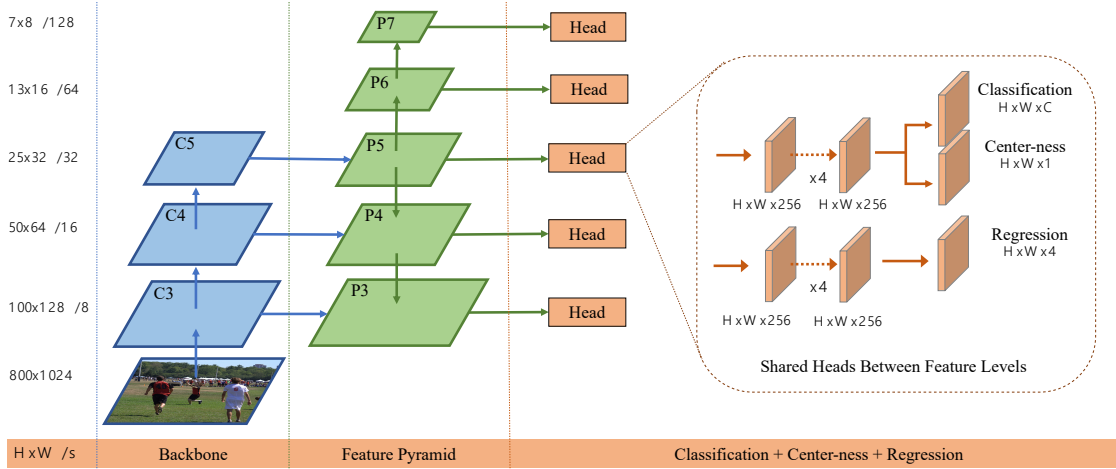


图2 – FCOS的网络架构，其中C3、C4和C5表示骨干网络的特征图，P3至P7是用于最终预测的特征层级。 $H \times W$ 是特征图的高度和宽度。‘ $/s$ ’ ( $s = 8, 16, \dots, 128$ ) 是该层级特征图相对于输入图像的下采样比例。举例而言，所有数值均基于  $800 \times 1024$  的输入计算得出。

需要仔细调整以获得良好的性能。除了上述描述锚框形状的超参数外，基于锚框的检测器还需要其他超参数来将每个锚框标记为正样本、忽略样本或负样本。在以往的研究中，通常利用锚框与真实框的交并比 (IOU) 来确定锚框的标签 (*e.g.*, 若其IOU处于 $[0.5, 1]$ 区间内则为正样本锚框)。这些超参数对最终精度有显著影响，且需要启发式调优。同时，这些超参数专属于检测任务，使得检测任务偏离了其他密集预测任务（如语义分割）所使用的简洁全卷积网络架构。

无锚点检测器。最流行的无锚点检测器可能是YOLOv1 [21]。YOLOv1不使用锚框，而是在物体中心附近的位置预测边界框。仅使用中心附近的点是因为它们被认为能够产生更高质量的检测。然而，由于只有中心附近的点被用于预测边界框，正如YOLOv2 [22]所述，YOLOv1存在召回率低的问题。因此，YOLOv2 [22]也采用了锚框。与YOLOv1相比，FCOS利用真实边界框内的所有点来预测边界框，并通过提出的“中心度”分支抑制低质量检测边界框。因此，如我们的实验所示，FCOS能够提供与基于锚点的检测器相当的召回率。

CornerNet [13] 是近期提出的一种单阶段无锚框

自由检测器，它检测边界框的一对角点并将它们分组以形成最终检测到的边界框。CornerNet需要更复杂的后处理来将属于同一实例的角点对分组。为此目的，学习了一个额外的距离度量。

另一类无锚检测器，如[32]，基于DenseBox[12]。这类检测器曾被认为不适用于通用目标检测，因为难以处理重叠边界框且召回率相对较低。在本研究中，我们证明通过多级FPN预测可以大幅缓解这两个问题。此外，结合我们提出的中心度分支，这种更简单的检测器甚至能比基于锚点的同类方法实现更好的检测性能。

### 3. 我们的方法

在本节中，我们首先以逐像素预测的方式重新定义目标检测。接着，我们展示了如何利用多层级预测来提高召回率，并解决由重叠边界框引起的歧义问题。最后，我们提出了所设计的“中心度”分支，该分支有助于抑制低质量的检测边界框，并大幅提升整体性能。

#### 3.1. 全卷积单阶段目标检测器

设  $F_i \in \mathbb{R}^{H \times W \times C}$  为骨干CNN第*i*层的特征图， $s$  为该层之前的总步长。输入图像的真实边界框为

defined as  $\{B_i\}$ , where  $B_i = (x_0^{(i)}, y_0^{(i)}, x_1^{(i)}, y_1^{(i)}, c^{(i)}) \in \mathbb{R}^4 \times \{1, 2 \dots C\}$ . Here  $(x_0^{(i)}, y_0^{(i)})$  and  $(x_1^{(i)}, y_1^{(i)})$  denote the coordinates of the left-top and right-bottom corners of the bounding box.  $c^{(i)}$  is the class that the object in the bounding box belongs to.  $C$  is the number of classes, which is 80 for MS-COCO dataset.

For each location  $(x, y)$  on the feature map  $F_i$ , we can map it back onto the input image as  $(\lfloor \frac{s}{2} \rfloor + xs, \lfloor \frac{s}{2} \rfloor + ys)$ , which is near the center of the receptive field of the location  $(x, y)$ . Different from anchor-based detectors, which consider the location on the input image as the center of (multiple) anchor boxes and regress the target bounding box with these anchor boxes as references, we directly regress the target bounding box at the location. In other words, our detector directly views locations as *training samples* instead of anchor boxes in anchor-based detectors, which is the same as FCNs for semantic segmentation [20].

Specifically, location  $(x, y)$  is considered as a positive sample if it falls into any ground-truth box and the class label  $c^*$  of the location is the class label of the ground-truth box. Otherwise it is a negative sample and  $c^* = 0$  (background class). Besides the label for classification, we also have a 4D real vector  $\mathbf{t}^* = (l^*, t^*, r^*, b^*)$  being the regression targets for the location. Here  $l^*, t^*, r^*$  and  $b^*$  are the distances from the location to the four sides of the bounding box, as shown in Fig. 1 (left). If a location falls into multiple bounding boxes, it is considered as an *ambiguous sample*. We simply choose the bounding box with minimal area as its regression target. In the next section, we will show that with multi-level prediction, the number of ambiguous samples can be reduced significantly and thus they hardly affect the detection performance. Formally, if location  $(x, y)$  is associated to a bounding box  $B_i$ , the training regression targets for the location can be formulated as,

$$\begin{aligned} l^* &= x - x_0^{(i)}, \quad t^* = y - y_0^{(i)}, \\ r^* &= x_1^{(i)} - x, \quad b^* = y_1^{(i)} - y. \end{aligned} \quad (1)$$

*It is worth noting that FCOS can leverage as many foreground samples as possible to train the regressor.* It is different from anchor-based detectors, which only consider the anchor boxes with a highly enough IOU with ground-truth boxes as positive samples. We argue that it may be one of the reasons that FCOS outperforms its anchor-based counterparts.

**Network Outputs.** Corresponding to the training targets, the final layer of our networks predicts an 80D vector  $\mathbf{p}$  of classification labels and a 4D vector  $\mathbf{t} = (l, t, r, b)$  bounding box coordinates. Following [15], instead of training a multi-class classifier, we train  $C$  binary classifiers. Similar to [15], we add four convolutional layers after the feature maps of the backbone networks respectively for clas-

sification and regression branches. Moreover, since the regression targets are always positive, we employ  $\exp(x)$  to map any real number to  $(0, \infty)$  on the top of the regression branch. *It is worth noting that FCOS has  $9\times$  fewer network output variables than the popular anchor-based detectors [15, 24] with 9 anchor boxes per location.*

**Loss Function.** We define our training loss function as follows:

$$\begin{aligned} L(\{\mathbf{p}_{x,y}\}, \{\mathbf{t}_{x,y}\}) &= \frac{1}{N_{\text{pos}}} \sum_{x,y} L_{\text{cls}}(\mathbf{p}_{x,y}, c_{x,y}^*) \\ &+ \frac{\lambda}{N_{\text{pos}}} \sum_{x,y} \mathbb{1}_{\{c_{x,y}^* > 0\}} L_{\text{reg}}(\mathbf{t}_{x,y}, \mathbf{t}_{x,y}^*), \end{aligned} \quad (2)$$

where  $L_{\text{cls}}$  is focal loss as in [15] and  $L_{\text{reg}}$  is the IOU loss as in UnitBox [32].  $N_{\text{pos}}$  denotes the number of positive samples and  $\lambda$  being 1 in this paper is the balance weight for  $L_{\text{reg}}$ . The summation is calculated over all locations on the feature maps  $F_i$ .  $\mathbb{1}_{\{c_i^* > 0\}}$  is the indicator function, being 1 if  $c_i^* > 0$  and 0 otherwise.

**Inference.** The inference of FCOS is straightforward. Given an input images, we forward it through the network and obtain the classification scores  $\mathbf{p}_{x,y}$  and the regression prediction  $\mathbf{t}_{x,y}$  for each location on the feature maps  $F_i$ . Following [15], we choose the location with  $p_{x,y} > 0.05$  as positive samples and invert Eq. (1) to obtain the predicted bounding boxes.

### 3.2. Multi-level Prediction with FPN for FCOS

Here we show that how two possible issues of the proposed FCOS can be resolved with multi-level prediction with FPN [14]. 1) The large stride (e.g.,  $16\times$ ) of the final feature maps in a CNN can result in a relatively low *best possible recall (BPR)*<sup>1</sup>. For anchor based detectors, low recall rates due to the large stride can be compensated to some extent by lowering the required IOU scores for positive anchor boxes. For FCOS, at the first glance one may think that the BPR can be much lower than anchor-based detectors because it is impossible to recall an object which no location on the final feature maps encodes due to a large stride. Here, we empirically show that even with a large stride, FCN-based FCOS is still able to produce a good BPR, and it can even better than the BPR of the anchor-based detector RetinaNet [15] in the official implementation Detectron [7] (refer to Table 1). Therefore, the BPR is actually not a problem of FCOS. Moreover, with multi-level FPN prediction [14], the BPR can be improved further to match the

<sup>1</sup>Upper bound of the recall rate that a detector can achieve.

定义为 $\{B_i\}$ , 其中 $B_i = (x_0^{(i)}, y_0^{(i)}, x_1^{(i)}, y_1^{(i)}, c^{(i)}) \in \mathbb{R}^4 \times \{1, 2 \dots C\}$ 。这里 $(x_0^{(i)}, y_0^{(i)})$ 和 $(x_1^{(i)}, y_1^{(i)})$ 表示边界框左上角和右下角的坐标。 $c^{(i)}$ 是边界框内物体所属的类别。 $C$ 是类别数量, 在MS-COCO数据集中为80。

对于特征图 $F_i$ 上的每个位置 $(x, y)$ , 我们可以将其映射回输入图像上的 $(\lfloor \frac{s}{2} \rfloor + xs, \lfloor \frac{s}{2} \rfloor + ys)$ , 该点位于该位置 $(x, y)$ 感受野的中心附近。不同于基于锚框的检测器——它们将输入图像上的位置视为(多个)锚框的中心, 并以这些锚框为参考回归目标边界框——我们直接在位置上回归目标边界框。换句话说, 我们的检测器直接将位置视为*training samples*, 而非基于锚框检测器中的锚框, 这与用于语义分割的FCNs做法一致[20]。

具体来说, 如果位置 $(x, y)$ 落入任意真实标注框内, 且该位置的类别标签 $c^*$ 与真实标注框的类别标签一致, 则将其视为正样本。否则, 它将被视为负样本, 且 $c^* = 0$  (背景类别)。除了分类标签外, 我们还有一个4维实向量 $\mathbf{t}^* = (l^*, t^*, r^*, b^*)$ 作为该位置的回归目标。其中 $l^*$ 、 $t^*$ 、 $r^*$ 和 $b^*$ 分别表示该位置到边界框四条边的距离, 如图1 (左) 所示。如果一个位置落入多个边界框内, 则将其视为*ambiguoussample*。我们直接选择面积最小的边界框作为其回归目标。在下一节中, 我们将展示通过多级预测, 模糊样本的数量可以显著减少, 因此它们几乎不会影响检测性能。形式上, 如果位置 $(x, y)$ 与边界框 $B_i$ 相关联, 则该位置的训练回归目标可表示为:

$$\begin{aligned} l^* &= x - x_0^{(i)}, \quad t^* = y - y_0^{(i)}, \\ r^* &= x_1^{(i)} - x, \quad b^* = y_1^{(i)} - y. \end{aligned} \quad (1)$$

*It is worth noting that FCOS can leverage as many foreground samples as possible to train the regressor.* 与基于锚点的检测器不同, 后者仅将与真实边界框具有足够高IOU的锚框视为正样本。我们认为, 这可能是FCOS 优于其基于锚点的对应方法的原因之一。

网络输出。与训练目标相对应, 我们网络的最后一层预测一个80维的分类标签向量 $\mathbf{p}$ 和一个4维的边界框坐标向量 $\mathbf{t} = (l, t, r, b)$ 。遵循[15]的方法, 我们训练 $C$ 个二元分类器, 而非训练一个多类分类器。与[15]类似, 我们在骨干网络的特征图后分别添加四个卷积层用于分类——

分类和回归分支。此外, 由于回归目标始终为正, 我们在回归分支的顶部采用 $\exp(x)$ 将任意实数映射到 $(0, \infty)$ 区间。*It is worth noting that FCOS has 9  $\times$  fewer network output variables than the popular anchor-based detectors [15, 24] with 9 anchor boxes per location.*

损失函数。我们将训练损失函数定义如下:

$$\begin{aligned} L(\{\mathbf{p}_{x,y}\}, \{\mathbf{t}_{x,y}\}) &= \frac{1}{N_{\text{pos}}} \sum_{x,y} L_{\text{cls}}(\mathbf{p}_{x,y}, c_{x,y}^*) \\ &+ \frac{\lambda}{N_{\text{pos}}} \sum_{x,y} \mathbb{1}_{\{c_{x,y}^* > 0\}} L_{\text{reg}}(\mathbf{t}_{x,y}, \mathbf{t}_{x,y}^*), \end{aligned} \quad (2)$$

其中 $L_{\text{cls}}$ 是如文献[15]所述的焦点损失,  $L_{\text{reg}}$ 是UnitBox [32] 中的IOU损失。 $N_{\text{pos}}$ 表示正样本的数量, 本文中 $\lambda$ 设为1, 作为 $L_{\text{reg}}$ 的平衡权重。求和计算在特征图 $F_i$ 的所有位置上进行。 $\mathbb{1}_{\{c_i^* > 0\}}$ 是指示函数, 当 $c_i^* > 0$ 时为1, 否则为0。

推理。FCOS的推理过程是直接的。给定输入图像, 我们通过网络前向传播, 获取特征图 $F_i$ 上每个位置的分类得分 $\mathbf{p}_{x,y}$ 和回归预测 $\mathbf{t}_{x,y}$ 。遵循[15]的方法, 我们选择 $p_{x,y} > 0.05$ 的位置作为正样本, 并通过反解公式(1)得到预测边界框。

### 3.2. 基于FPN的FCOS多级预测

在此我们展示了如何通过使用FPN[14]的多层预测来解决所提出的FCOS可能存在的两个问题。1) CNN中最终特征图的大步长(e.g.,  $16\times$ )可能导致相对较低的*best possible recall (BPR)*<sup>1</sup>。对于基于锚点的检测器, 大步长导致的低召回率可以通过降低正锚框所需的IOU分数在一定程度上得到补偿。对于FCOS, 乍看之下人们可能会认为其BPR可能远低于基于锚点的检测器, 因为由于大步长, 最终特征图上没有位置编码的对象是无法被召回的。然而, 我们通过实验证明, 即使在大步长的情况下, 基于FCN的FCOS仍能产生良好的BPR, 甚至在官方实现Detectron[7]中, 其BPR可以优于基于锚点的检测器RetinaNet[15] (参见表1)。因此, BPR实际上并非FCOS的问题。此外, 通过多层FPN预测[14], BPR可以进一步提升以达到匹配效果。

<sup>1</sup>Upper bound of the recall rate that a detector can achieve.



best BPR the anchor-based RetinaNet can achieve. 2) Overlaps in ground-truth boxes can cause intractable ambiguity, *i.e.*, which bounding box should a location in the overlap regress? This ambiguity results in degraded performance of FCN-based detectors. In this work, we show that the ambiguity can be greatly resolved with multi-level prediction, and the FCN-based detector can obtain *on par*, sometimes even better, performance compared with anchor-based ones.

Following FPN [14], we detect different sizes of objects on different levels of feature maps. Specifically, we make use of five levels of feature maps defined as  $\{P_3, P_4, P_5, P_6, P_7\}$ .  $P_3, P_4$  and  $P_5$  are produced by the backbone CNNs' feature maps  $C_3, C_4$  and  $C_5$  followed by a  $1 \times 1$  convolutional layer with the top-down connections in [14], as shown in Fig. 2.  $P_6$  and  $P_7$  are produced by applying one convolutional layer with the stride being 2 on  $P_5$  and  $P_6$ , respectively. As a result, the feature levels  $P_3, P_4, P_5, P_6$  and  $P_7$  have strides 8, 16, 32, 64 and 128, respectively.

Unlike anchor-based detectors, which assign anchor boxes with different sizes to different feature levels, we directly limit the range of bounding box regression for each level. More specifically, we firstly compute the regression targets  $l^*, t^*, r^*$  and  $b^*$  for each location on all feature levels. Next, if a location satisfies  $\max(l^*, t^*, r^*, b^*) > m_i$  or  $\max(l^*, t^*, r^*, b^*) < m_{i-1}$ , it is set as a negative sample and is thus not required to regress a bounding box anymore. Here  $m_i$  is the maximum distance that feature level  $i$  needs to regress. In this work,  $m_2, m_3, m_4, m_5, m_6$  and  $m_7$  are set as 0, 64, 128, 256, 512 and  $\infty$ , respectively. Since objects with different sizes are assigned to different feature levels and most overlapping happens between objects with considerably different sizes. If a location, even with multi-level prediction used, is still assigned to more than one ground-truth boxes, we simply choose the ground-truth box with minimal area as its target. As shown in our experiments, the multi-level prediction can largely alleviate the aforementioned ambiguity and improve the FCN-based detector to the same level of anchor-based ones.

Finally, following [14, 15], we share the heads between different feature levels, not only making the detector parameter-efficient but also improving the detection performance. However, we observe that different feature levels are required to regress different size range (*e.g.*, the size range is  $[0, 64]$  for  $P_3$  and  $[64, 128]$  for  $P_4$ ), and therefore it is not reasonable to make use of identical heads for different feature levels. As a result, instead of using the standard  $\exp(x)$ , we make use of  $\exp(s_i x)$  with a trainable scalar  $s_i$  to automatically adjust the base of the exponential function for feature level  $P_i$ , which slightly improves the detection performance.

### 3.3. Center-ness for FCOS

After using multi-level prediction in FCOS, there is still a performance gap between FCOS and anchor-based detectors. We observed that it is due to a lot of low-quality predicted bounding boxes produced by locations far away from the center of an object.

We propose a simple yet effective strategy to suppress these low-quality detected bounding boxes without introducing any hyper-parameters. Specifically, we add a *single-layer branch*, in parallel with the classification branch (as shown in Fig. 2) to predict the “center-ness” of a location<sup>2</sup>. The center-ness depicts the normalized distance from the location to the center of the object that the location is responsible for, as shown Fig. 7. Given the regression targets  $l^*, t^*, r^*$  and  $b^*$  for a location, the center-ness target is defined as,

$$\text{centerness}^* = \sqrt{\frac{\min(l^*, r^*)}{\max(l^*, r^*)} \times \frac{\min(t^*, b^*)}{\max(t^*, b^*)}}. \quad (3)$$

We employ sqrt here to slow down the decay of the center-ness. The center-ness ranges from 0 to 1 and is thus trained with binary cross entropy (BCE) loss. The loss is added to the loss function Eq. (2). When testing, the final score (used for ranking the detected bounding boxes) is computed by multiplying the predicted center-ness with the corresponding classification score. Thus the center-ness can down-weight the scores of bounding boxes far from the center of an object. As a result, with high probability, these low-quality bounding boxes might be filtered out by the final non-maximum suppression (NMS) process, improving the detection performance *remarkably*.

An alternative of the center-ness is to make use of only the central portion of ground-truth bounding box as positive samples with the price of one extra hyper-parameter, as shown in works [12, 33]. After our submission, it has been shown in [1] that the combination of both methods can achieve a much better performance. The experimental results can be found in Table 3.

## 4. Experiments

Our experiments are conducted on the large-scale detection benchmark COCO [16]. Following the common practice [15, 14, 24], we use the COCO `trainval35k` split (115K images) for training and `minival` split (5K images) as validation for our ablation study. We report our main results on the `test_dev` split (20K images) by uploading our detection results to the evaluation server.

<sup>2</sup>After the initial submission, it has been shown that the AP on MS-COCO can be improved if the center-ness is parallel with the regression branch instead of the classification branch. However, unless specified, we still use the configuration in Fig. 2.

基于锚点的RetinaNet所能达到的最佳BPR。2) 真实标注框之间的重叠会导致难以处理的歧义问题, *i.e.*, 即重叠区域内的位置应该回归到哪个边界框? 这种歧义导致基于FCN的检测器性能下降。在本研究中, 我们证明通过多层级预测可以极大程度地解决这种歧义, 并且基于FCN的检测器能够取得*on par*, 有时甚至优于基于锚点的检测器的性能。

遵循FPN[14]的方法, 我们在不同层级的特征图上检测不同尺寸的目标。具体而言, 我们利用五个层级的特征图, 定义为 $\{P_3, P_4, P_5, P_6, P_7\}$ 。  $P_3$ 、 $P_4$ 和 $P_5$ 由骨干CNN的特征图 $C_3$ 、 $C_4$ 和 $C_5$ 通过[14]中所述的自顶向下连接并接一个 $1 \times 1$ 卷积层生成, 如图2所示。  $P_6$ 和 $P_7$ 则分别通过对 $P_5$ 和 $P_6$ 应用步长为2的卷积层得到。最终, 特征层级 $P_3$ 、 $P_4$ 、 $P_5$ 、 $P_6$ 和 $P_7$ 的步长分别为8、16、32、64和128。

与基于锚点的检测器将不同尺寸的锚框分配给不同特征层不同, 我们直接限制每个层级的边界框回归范围。具体而言, 我们首先为所有特征层上的每个位置计算回归目标 $l^*$ 、 $t^*$ 、 $r^*$ 和 $b^*$ 。接着, 若某位置满足 $\max(l^*, t^*, r^*, b^*) > m_i$ 或 $\max(l^*, t^*, r^*, b^*) < m_{i-1}$ , 则将其设为负样本, 不再需要进行边界框回归。此处 $m_i$ 是特征层 $i$ 所需回归的最大距离。本研究中,  $m_2$ 、 $m_3$ 、 $m_4$ 、 $m_5$ 、 $m_6$ 和 $m_7$ 分别设置为0、64、128、256、512和 $\infty$ 。由于不同尺寸的目标被分配到不同特征层, 且重叠主要发生在尺寸差异显著的目标之间。若某位置即使采用多层级预测仍被分配给多个真实框, 我们直接选择面积最小的真实框作为其目标。实验表明, 多层级预测能显著缓解前述歧义, 并将基于全卷积网络的检测器提升至与基于锚点的检测器相当的水平。

最后, 遵循[14, 15]的做法, 我们在不同特征层级间共享检测头, 这不仅使检测器参数高效, 还提升了检测性能。然而, 我们观察到不同特征层级需要回归不同尺寸范围 (*e.g.*,  $P_3$ 的尺寸范围为[0,64],  $P_4$ 的尺寸范围为[64,128]), 因此为不同特征层级使用相同的检测头并不合理。于是, 我们不再使用标准的 $\exp(x)$ , 而是采用 $\exp(s_i x)$ 并引入可训练标量 $s_i$ , 以自动为特征层级 $P_i$ 调整指数函数的底数, 这略微提升了检测性能。

### 3.3. FCOS的中心度

在使用FCOS进行多级预测后, FCOS与基于锚点的检测器之间仍存在性能差距。我们观察到, 这是由于距离物体中心较远的位置产生了大量低质量的预测边界框。

我们提出了一种简单而有效的策略来抑制这些低质量的检测边界框, 且无需引入任何超参数。具体而言, 我们在分类分支之外并行添加了 $a$  *single-layer branch*, 以预测位置<sup>2</sup>的“中心度”。如图7所示, 中心度描述了该位置到其负责的目标中心的归一化距离。给定位置对应的回归目标 $l^*$ 、 $t^*$ 、 $r^*$ 和 $b^*$ , 中心度目标定义为:

$$\text{centerness}^* = \sqrt{\frac{\min(l^*, r^*)}{\max(l^*, r^*)} \times \frac{\min(t^*, b^*)}{\max(t^*, b^*)}}. \quad (3)$$

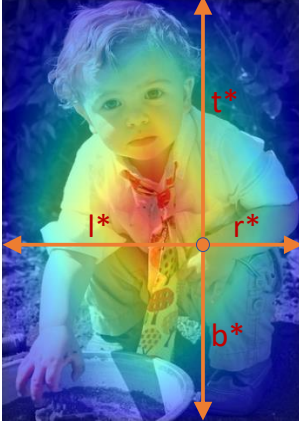
我们在此采用平方根运算以减缓中心度的衰减。中心度的取值范围为0到1, 因此使用二元交叉熵 (BCE) 损失进行训练。该损失被加入损失函数公式 (2) 中。在测试时, 最终得分 (用于对检测到的边界框进行排序) 通过将预测的中心度与相应的分类得分相乘来计算。因此, 中心度可以降低远离物体中心的边界框的得分权重。这样一来, 这些低质量的边界框有很大概率会在最终的非极大值抑制 (NMS) 过程中被过滤掉, 从而提升检测性能*remarkably*。

中心度的一种替代方法是仅使用真实边界框的中心部分作为正样本, 代价是增加一个额外的超参数, 如文献[12, 33]所示。在我们提交后, 文献[1]表明两种方法的结合可以实现更好的性能。实验结果见表3。

## 4. 实验

我们的实验在大规模检测基准COCO[16]上进行。遵循通用实践[15, 14, 24], 我们使用COCO trainval35k划分 (11.5万张图像) 进行训练, 并采用minival划分 (5千张图像) 作为验证集以开展消融研究。我们将检测结果提交至评估服务器, 在test\_dev划分 (2万张图像) 上报告了主要实验结果。

<sup>2</sup>After the initial submission, it has been shown that the AP on MS-COCO can be improved if the center-ness is parallel with the regression branch instead of the classification branch. However, unless specified, we still use the configuration in Fig. 2.



**Figure 3 – Center-ness.** Red, blue, and other colors denote 1, 0 and the values between them, respectively. Center-ness is computed by Eq. (3) and decays from 1 to 0 as the location deviates from the center of the object. When testing, the center-ness predicted by the network is multiplied with the classification score thus can down-weight the low-quality bounding boxes predicted by a location far from the center of an object.

**Training Details.** Unless specified, ResNet-50 [8] is used as our backbone networks and the same hyper-parameters with RetinaNet [15] are used. Specifically, our network is trained with stochastic gradient descent (SGD) for 90K iterations with the initial learning rate being 0.01 and a mini-batch of 16 images. The learning rate is reduced by a factor of 10 at iteration 60K and 80K, respectively. Weight decay and momentum are set as 0.0001 and 0.9, respectively. We initialize our backbone networks with the weights pre-trained on ImageNet [4]. For the newly added layers, we initialize them as in [15]. Unless specified, the input images are resized to have their shorter side being 800 and their longer side less or equal to 1333.

**Inference Details.** We firstly forward the input image through the network and obtain the predicted bounding boxes with a predicted class. Unless specified, the following post-processing is exactly the same with RetinaNet [15] and we directly make use of the same post-processing hyper-parameters of RetinaNet. We use the same sizes of input images as in training. We hypothesize that the performance of our detector may be improved further if we carefully tune the hyper-parameters.

## 4.1. Ablation Study

### 4.1.1 Multi-level Prediction with FPN

As mentioned before, the major concerns of an FCN-based detector are *low recall* rates and *ambiguous samples* resulted from overlapping in ground-truth bounding boxes. In the section, we show that both issues can be largely resolved

| Method    | w/ FPN | Low-quality matches | BPR (%)      |
|-----------|--------|---------------------|--------------|
| RetinaNet | ✓      | None                | 86.82        |
| RetinaNet | ✓      | $\geq 0.4$          | 90.92        |
| RetinaNet | ✓      | All                 | <b>99.23</b> |
| FCOS      |        | -                   | 95.55        |
| FCOS      | ✓      | -                   | 98.40        |

**Table 1** – The BPR for anchor-based RetinaNet under a variety of matching rules and the BPR for FCN-based FCOS. FCN-based FCOS has very similar recall to the best anchor-based one and has much higher recall than the official implementation in Detectron [7], where only low-quality matches with  $\text{IOU} \geq 0.4$  are considered.

| w/ FPN | Amb. samples (%) | Amb. samples (diff.) (%) |
|--------|------------------|--------------------------|
|        | 23.16            | 17.84                    |
| ✓      | <b>7.14</b>      | <b>3.75</b>              |

**Table 2** – Amb. samples denotes the ratio of the ambiguous samples to all positive samples. Amb. samples (diff.) is similar but excludes those ambiguous samples in the overlapped regions but belonging to the same category as the kind of ambiguity does not matter when inferring. We can see that with FPN, this percentage of ambiguous samples is small (3.75%).

with multi-level prediction.

**Best Possible Recalls.** The first concern about the FCN-based detector is that it might not provide a good best possible recall (BPR). In the section, we show that the concern is not necessary. Here BPR is defined as the ratio of the number of ground-truth boxes a detector can recall at the most divided by all ground-truth boxes. A ground-truth box is considered being recalled if the box is assigned to at least one sample (*i.e.*, a location in FCOS or an anchor box in anchor-based detectors) during training. As shown in Table 1, only with feature level  $P_4$  with stride being 16 (*i.e.*, no FPN), FCOS can already obtain a BPR of 95.55%. The BPR is much higher than the BPR of 90.92% of the anchor-based detector RetinaNet in the official implementation Detectron, where only the low-quality matches with  $\text{IOU} \geq 0.4$  are used. With the help of FPN, FCOS can achieve a BPR of 98.40%, which is very close to the best BPR that the anchor-based detector can achieve by using all low-quality matches. Due to the fact that the best recall of current detectors are much lower than 90%, the small BPR gap (less than 1%) between FCOS and the anchor-based detector will not actually affect the performance of detector. It is also confirmed in Table 3, where FCOS achieves even better AR than its anchor-based counterparts under the same training and testing settings. Therefore, the concern about low BPR may not be necessary.

**Ambiguous Samples.** Another concern about the FCN-based detector is that it may have a large number of *ambigu-*

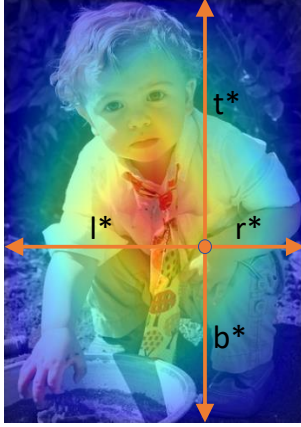


图3 – 中心度。红色、蓝色及其他颜色分别表示1、0及介于两者之间的数值。中心度通过公式(3)计算，并随着位置偏离物体中心从1衰减至0。测试时，网络预测的中心度会与分类得分相乘，从而降低远离物体中心的位置所预测的低质量边界框的权重。

训练细节。除非特别说明，我们使用ResNet-50 [8]作为骨干网络，并采用与RetinaNet [15]相同的超参数。具体而言，我们的网络使用随机梯度下降（SGD）进行训练，共90K次迭代，初始学习率为0.01，小批量大小为16张图像。学习率在迭代60K和80K时分别降低10倍。权重衰减和动量分别设置为0.0001和0.9。我们使用在ImageNet [4]上预训练的权重初始化骨干网络。对于新增的层，我们按照[15]的方法进行初始化。除非特别说明，输入图像的短边调整为800像素，长边小于或等于1333像素。

推理细节。我们首先将输入图像通过网络前向传播，获得带有预测类别的边界框预测。除非另有说明，以下后处理步骤与RetinaNet[15]完全相同，我们直接沿用RetinaNet相同的后处理超参数。我们使用与训练时相同尺寸的输入图像。我们推测，若仔细调整超参数，检测器的性能可能得到进一步提升。

## 4.1. 消融研究

### 4.1.1 基于FPN的多级预测

如前所述，基于FCN的检测器主要关注的是由真实边界框重叠引起的*low recall*率和*ambiguous samples*问题。在本节中，我们将展示这两个问题在很大程度上可以得到解决。

| Method    | w/ FPN | Low-quality matches | BPR (%)      |
|-----------|--------|---------------------|--------------|
| RetinaNet | ✓      | None                | 86.82        |
| RetinaNet | ✓      | $\geq 0.4$          | 90.92        |
| RetinaNet | ✓      | All                 | <b>99.23</b> |
| FCOS      |        | -                   | 95.55        |
| FCOS      | ✓      | -                   | 98.40        |

表1 – 基于锚点的RetinaNet在不同匹配规则下的BPR，以及基于FCN的FCOS的BPR。基于FCN的FCOS的召回率与最佳的基于锚点的方法非常接近，且远高于Detectron官方实现[7]中的召回率——后者仅考虑 $\text{IOU} \geq 0.4$ 的低质量匹配。

| w/ FPN | Amb. samples (%) | Amb. samples (diff.) (%) |
|--------|------------------|--------------------------|
|        | 23.16            | 17.84                    |
| ✓      | <b>7.14</b>      | <b>3.75</b>              |

表2 – Amb. samples 表示模糊样本占有所有正样本的比例。Amb. samples (diff.) 与之类似，但排除了重叠区域中属于同一类别的模糊样本，因为在推断时这类模糊性并不重要。可以看出，在使用FPN的情况下，这类模糊样本的占比比较小（3.75%）。

采用多级预测。

最佳可能召回率。关于基于FCN的检测器的首要担忧是，它可能无法提供良好的最佳可能召回率（BPR）。在本节中，我们表明这种担忧是不必要的。这里BPR定义为检测器最多能召回的真实标注框数量除以所有真实标注框的比率。如果一个真实标注框在训练期间被分配给至少一个样本（*i.e.*，例如FCOS中的某个位置或基于锚框检测器中的锚框），则认为该框被召回。如表1所示，仅使用步长为16的特征层级 $P_4$ （*i.e.*，即不使用FPN），FCOS已经可以获得95.55%的BPR。这一BPR远高于官方实现Detectron中基于锚框的检测器RetinaNet的90.92% BPR，后者仅使用了 $\text{IOU} \geq 0.4$ 的低质量匹配。借助FPN，FCOS可以实现98.40%的BPR，非常接近基于锚框的检测器通过使用所有低质量匹配所能达到的最佳BPR。由于当前检测器的最佳召回率远低于90%，FCOS与基于锚框检测器之间的小幅BPR差距（小于1%）实际上不会影响检测器的性能。表3也证实了这一点，在相同的训练和测试设置下，FCOS甚至实现了比基于锚框的对应方法更好的平均召回率（AR）。因此，对低BPR的担忧可能是不必要的。

模糊样本。关于基于FCN的检测器的另一个担忧是，它可能包含大量*ambigu-*

| Method              | $C_5/P_5$ | w/ GN | nms thr. | AP          | AP <sub>50</sub> | AP <sub>75</sub> | AP <sub>S</sub> | AP <sub>M</sub> | AP <sub>L</sub> | AR <sub>1</sub> | AR <sub>10</sub> | AR <sub>100</sub> |
|---------------------|-----------|-------|----------|-------------|------------------|------------------|-----------------|-----------------|-----------------|-----------------|------------------|-------------------|
| RetinaNet           | $C_5$     |       | .50      | 35.9        | 56.0             | 38.2             | 20.0            | 39.8            | 47.4            | 31.0            | 49.4             | 52.5              |
| FCOS                | $C_5$     |       | .50      | 36.3        | 54.8             | 38.7             | 20.5            | 39.8            | 47.8            | 31.5            | 50.6             | 53.5              |
| FCOS                | $P_5$     |       | .50      | 36.4        | 54.9             | 38.8             | 19.7            | 39.7            | 48.8            | 31.4            | 50.6             | 53.4              |
| FCOS                | $P_5$     |       | .60      | 36.5        | 54.5             | 39.2             | 19.8            | 40.0            | 48.9            | 31.3            | 51.2             | 54.5              |
| FCOS                | $P_5$     | ✓     | .60      | 37.1        | 55.9             | 39.8             | 21.3            | 41.0            | 47.8            | 31.4            | 51.4             | 54.9              |
| <b>Improvements</b> |           |       |          |             |                  |                  |                 |                 |                 |                 |                  |                   |
| + ctr. on reg.      | $P_5$     | ✓     | .60      | 37.4        | 56.1             | 40.3             | 21.8            | 41.2            | 48.8            | 31.5            | 51.7             | 55.2              |
| + ctr. sampling [1] | $P_5$     | ✓     | .60      | 38.1        | 56.7             | <b>41.4</b>      | <b>22.6</b>     | 41.6            | <b>50.4</b>     | 32.1            | 52.8             | 56.3              |
| + GIoU [1]          | $P_5$     | ✓     | .60      | 38.3        | 57.1             | 41.0             | 21.9            | 42.4            | 49.5            | 32.0            | 52.9             | 56.5              |
| + Normalization     | $P_5$     | ✓     | .60      | <b>38.6</b> | <b>57.4</b>      | <b>41.4</b>      | 22.3            | <b>42.5</b>     | 49.8            | <b>32.3</b>     | <b>53.4</b>      | <b>57.1</b>       |

**Table 3** – FCOS vs. RetinaNet on the `minival` split with ResNet-50-FPN as the backbone. Directly using the training and testing settings of RetinaNet, our anchor-free FCOS achieves even better performance than anchor-based RetinaNet both in AP and AR. With Group Normalization (GN) in heads and NMS threshold being 0.6, FCOS can achieve 37.1 in AP. After our submission, some almost cost-free improvements have been made for FCOS and the performance has been improved by a large margin, as shown by the rows below “**Improvements**”. “ctr. on reg.”: moving the center-ness branch to the regression branch. “ctr. sampling”: only sampling the central portion of ground-truth boxes as positive samples. “GIoU”: penalizing the union area over the circumscribed rectangle’s area in IoU Loss. “Normalization”: normalizing the regression targets in Eq. (1) with the strides of FPN levels. Refer to our code for details.

ous samples due to the overlapping in ground-truth bounding boxes, as shown in Fig. 1 (right). In Table 2, we show the ratios of the ambiguous samples to all positive samples on `minival` split. As shown in the table, there are indeed a large amount of ambiguous samples (23.16%) if FPN is not used and only feature level  $P_4$  is used. However, with FPN, the ratio can be significantly reduced to only 7.14% since most of overlapped objects are assigned to different feature levels. Moreover, we argue that the ambiguous samples resulted from overlapping between objects of the same category do not matter. For instance, if object A and B with the same class have overlap, no matter which object the locations in the overlap predict, the prediction is correct because it is always matched with the same category. The missed object can be predicted by the locations only belonging to it. Therefore, we only count the ambiguous samples in overlap between bounding boxes with different categories. As shown in Table 2, the multi-level prediction reduces the ratio of ambiguous samples from 17.84% to 3.75%. In order to further show that the overlapping in ground truth boxes is not a problem of our FCN-based FCOS, we count that when inferring how many detected bounding boxes come from the ambiguous locations. We found that only 2.3% detected bounding boxes are produced by the ambiguous locations. By further only considering the overlap between different categories, the ratio is reduced to 1.5%. Note that it does not imply that there are 1.5% locations where FCOS cannot work. As mentioned before, these locations are associated with the ground-truth boxes with minimal area. Therefore, these locations only take the risk of missing some larger objects. As shown in the following experiments, they do not make our FCOS inferior to anchor-based detectors.

|                          | AP          | AP <sub>50</sub> | AP <sub>75</sub> | AP <sub>S</sub> | AP <sub>M</sub> | AP <sub>L</sub> |
|--------------------------|-------------|------------------|------------------|-----------------|-----------------|-----------------|
| None                     | 33.5        | 52.6             | 35.2             | 20.8            | 38.5            | 42.6            |
| center-ness <sup>†</sup> | 33.5        | 52.4             | 35.1             | 20.8            | 37.8            | 42.8            |
| center-ness              | <b>37.1</b> | <b>55.9</b>      | <b>39.8</b>      | <b>21.3</b>     | <b>41.0</b>     | <b>47.8</b>     |

**Table 4** – Ablation study for the proposed center-ness branch on `minival` split. “None” denotes that no center-ness is used. “center-ness<sup>†</sup>” denotes that using the center-ness computed from the predicted regression vector. “center-ness” is that using center-ness predicted from the proposed center-ness branch. The center-ness branch improves the detection performance under all metrics.

#### 4.1.2 With or Without Center-ness

As mentioned before, we propose “center-ness” to suppress the low-quality detected bounding boxes produced by the locations far from the center of an object. As shown in Table 4, the center-ness branch can boost AP from 33.5% to 37.1%, making anchor-free FCOS outperform anchor-based RetinaNet (35.9%). Note that anchor-based RetinaNet employs two IoU thresholds to label anchor boxes as positive/negative samples, which can also help to suppress the low-quality predictions. The proposed center-ness can eliminate the two hyper-parameters. However, after our initial submission, it has shown that using both center-ness and the thresholds can result in a better performance, as shown by the row “+ ctr. sampling” in Table 3. One may note that center-ness can also be computed with the predicted regression vector without introducing the extra center-ness branch. However, as shown in Table 4, the center-ness computed from the regression vector cannot improve the performance and thus the separate center-ness is necessary.

#### 4.1.3 FCOS vs. Anchor-based Detectors

The aforementioned FCOS has two minor differences from the standard RetinaNet. 1) We use Group Normalization



| Method              | $C_5/P_5$ | w/ GN | nms thr. | AP          | AP <sub>50</sub> | AP <sub>75</sub> | AP <sub>S</sub> | AP <sub>M</sub> | AP <sub>L</sub> | AR <sub>1</sub> | AR <sub>10</sub> | AR <sub>100</sub> |
|---------------------|-----------|-------|----------|-------------|------------------|------------------|-----------------|-----------------|-----------------|-----------------|------------------|-------------------|
| RetinaNet           | $C_5$     |       | .50      | 35.9        | 56.0             | 38.2             | 20.0            | 39.8            | 47.4            | 31.0            | 49.4             | 52.5              |
| FCOS                | $C_5$     |       | .50      | 36.3        | 54.8             | 38.7             | 20.5            | 39.8            | 47.8            | 31.5            | 50.6             | 53.5              |
| FCOS                | $P_5$     |       | .50      | 36.4        | 54.9             | 38.8             | 19.7            | 39.7            | 48.8            | 31.4            | 50.6             | 53.4              |
| FCOS                | $P_5$     |       | .60      | 36.5        | 54.5             | 39.2             | 19.8            | 40.0            | 48.9            | 31.3            | 51.2             | 54.5              |
| FCOS                | $P_5$     | ✓     | .60      | 37.1        | 55.9             | 39.8             | 21.3            | 41.0            | 47.8            | 31.4            | 51.4             | 54.9              |
| <b>Improvements</b> |           |       |          |             |                  |                  |                 |                 |                 |                 |                  |                   |
| + ctr. on reg.      | $P_5$     | ✓     | .60      | 37.4        | 56.1             | 40.3             | 21.8            | 41.2            | 48.8            | 31.5            | 51.7             | 55.2              |
| + ctr. sampling [1] | $P_5$     | ✓     | .60      | 38.1        | 56.7             | <b>41.4</b>      | <b>22.6</b>     | 41.6            | <b>50.4</b>     | 32.1            | 52.8             | 56.3              |
| + GIoU [1]          | $P_5$     | ✓     | .60      | 38.3        | 57.1             | 41.0             | 21.9            | 42.4            | 49.5            | 32.0            | 52.9             | 56.5              |
| + Normalization     | $P_5$     | ✓     | .60      | <b>38.6</b> | <b>57.4</b>      | <b>41.4</b>      | 22.3            | <b>42.5</b>     | 49.8            | <b>32.3</b>     | <b>53.4</b>      | <b>57.1</b>       |

表3 – 以 ResNet-50-FPN 为骨干网络时，FCOS 与 RetinaNet 在 minival 数据集上的对比。直接采用 RetinaNet 的训练和测试设置，我们无锚框的 FCOS 在 AP 和 AR 上均取得了优于基于锚框的 RetinaNet 的性能。在检测头中使用组归一化（GN）且 NMS 阈值为 0.6 时，FCOS 的 AP 可达 37.1。在我们提交结果后，FCOS 进行了一些几乎无需额外成本的改进，性能得到大幅提升，如“改进项”下方各行所示。“ctr. on reg.”：将中心度分支移至回归分支。“ctr. sampling”：仅将真实框的中心区域采样为正样本。“GIoU”：在 IoU 损失中对外接矩形面积与并集面积的比值进行惩罚。“Normalization”：使用 FPN 层级的步长对公式（1）中的回归目标进行归一化。详见代码实现。

ous samples 由于真实边界框的重叠，如图1（右）所示。在表2中，我们展示了在minival划分上模糊样本占所有正样本的比例。如表所示，如果不使用FPN而仅使用特征层级 $P_4$ ，确实存在大量模糊样本（23.16%）。然而，使用FPN后，该比例可显著降低至仅7.14%，因为大多数重叠物体被分配到了不同的特征层级。此外，我们认为由同一类别物体间重叠导致的模糊样本并不重要。例如，若物体A和B属于同一类别且存在重叠，则无论重叠区域内的位置预测哪个物体，预测结果都是正确的，因为它始终与同一类别匹配。未被预测到的物体可通过仅属于其自身的位置进行预测。因此，我们仅统计不同类别边界框重叠产生的模糊样本。如表2所示，多层级预测将模糊样本的比例从17.84%降低至3.75%。为进一步证明真实边界框的重叠并非我们基于FCN的FCOS的问题，我们统计了推断时有多少检测到的边界框来源于模糊位置。我们发现仅2.3%的检测边界框由模糊位置产生。若进一步仅考虑不同类别间的重叠，该比例降至1.5%。需注意，这并不意味着存在1.5%的位置是FCOS无法处理的。如前所述，这些位置与面积最小的真实边界框相关联。因此，这些位置仅承担遗漏某些较大物体的风险。如下述实验所示，它们并未使我们的FCOS逊色于基于锚框的检测器。

|                          | AP          | AP <sub>50</sub> | AP <sub>75</sub> | AP <sub>S</sub> | AP <sub>M</sub> | AP <sub>L</sub> |
|--------------------------|-------------|------------------|------------------|-----------------|-----------------|-----------------|
| None                     | 33.5        | 52.6             | 35.2             | 20.8            | 38.5            | 42.6            |
| center-ness <sup>†</sup> | 33.5        | 52.4             | 35.1             | 20.8            | 37.8            | 42.8            |
| center-ness              | <b>37.1</b> | <b>55.9</b>      | <b>39.8</b>      | <b>21.3</b>     | <b>41.0</b>     | <b>47.8</b>     |

表4 – 在minival划分上对所提出的中心度分支进行的消融研究。“无”表示未使用任何中心度。“中心度<sup>†</sup>”表示使用从预测回归向量计算出的中心度。“中心度”表示使用从所提出的中心度分支预测的中心度。该中心度分支在所有指标下均提升了检测性能。

#### 4.1.2 带或不带中心度

如前所述，我们提出“中心度”来抑制那些远离物体中心位置所产生的低质量检测边界框。如表4所示，中心度分支能将AP从33.5%提升至37.1%，使得无锚框的FCOS超越基于锚框的RetinaNet（35.9%）。需要注意的是，基于锚框的RetinaNet采用两个IoU阈值将锚框标记为正/负样本，这同样有助于抑制低质量预测。我们提出的中心度方法可以消除这两个超参数。然而，在我们初次提交后，研究表明同时使用中心度和阈值能获得更好的性能，如表3中“+ ctr. sampling”行所示。有人可能注意到，中心度也可以通过预测的回归向量计算，无需引入额外的中心度分支。但如表4所示，基于回归向量计算的中心度无法提升性能，因此独立的中心度分支是必要的。

#### 4.1.3 FCOS与基于锚框的检测器对比

上述FCOS与标准RetinaNet有两个细微差别。1) 我们使用组归一化

| Method                     | Backbone                 | AP          | AP <sub>50</sub> | AP <sub>75</sub> | AP <sub>S</sub> | AP <sub>M</sub> | AP <sub>L</sub> |
|----------------------------|--------------------------|-------------|------------------|------------------|-----------------|-----------------|-----------------|
| Two-stage methods:         |                          |             |                  |                  |                 |                 |                 |
| Faster R-CNN w/ FPN [14]   | ResNet-101-FPN           | 36.2        | 59.1             | 39.0             | 18.2            | 39.0            | 48.2            |
| Faster R-CNN by G-RMI [11] | Inception-ResNet-v2 [27] | 34.7        | 55.5             | 36.7             | 13.5            | 38.1            | 52.0            |
| Faster R-CNN w/ TDM [25]   | Inception-ResNet-v2-TDM  | 36.8        | 57.7             | 39.2             | 16.2            | 39.8            | 52.1            |
| One-stage methods:         |                          |             |                  |                  |                 |                 |                 |
| YOLOv2 [22]                | DarkNet-19 [22]          | 21.6        | 44.0             | 19.2             | 5.0             | 22.4            | 35.5            |
| SSD513 [18]                | ResNet-101-SSD           | 31.2        | 50.4             | 33.3             | 10.2            | 34.5            | 49.8            |
| DSSD513 [5]                | ResNet-101-DSSD          | 33.2        | 53.3             | 35.2             | 13.0            | 35.4            | 51.1            |
| RetinaNet [15]             | ResNet-101-FPN           | 39.1        | 59.1             | 42.3             | 21.8            | 42.7            | 50.2            |
| CornerNet [13]             | Hourglass-104            | 40.5        | 56.5             | 43.1             | 19.4            | 42.7            | 53.9            |
| FSAF [34]                  | ResNeXt-64x4d-101-FPN    | 42.9        | 63.8             | 46.3             | 26.6            | 46.2            | 52.7            |
| FCOS                       | ResNet-101-FPN           | 41.5        | 60.7             | 45.0             | 24.4            | 44.8            | 51.6            |
| FCOS                       | HRNet-W32-51 [26]        | 42.0        | 60.4             | 45.3             | 25.4            | 45.0            | 51.0            |
| FCOS                       | ResNeXt-32x8d-101-FPN    | 42.7        | 62.2             | 46.1             | 26.0            | 45.6            | 52.6            |
| FCOS                       | ResNeXt-64x4d-101-FPN    | 43.2        | 62.8             | 46.6             | 26.5            | 46.2            | 53.3            |
| FCOS w/ improvements       | ResNeXt-64x4d-101-FPN    | <b>44.7</b> | <b>64.1</b>      | <b>48.4</b>      | <b>27.6</b>     | <b>47.5</b>     | <b>55.6</b>     |

**Table 5** – FCOS vs. other state-of-the-art two-stage or one-stage detectors (*single-model and single-scale results*). FCOS outperforms the anchor-based counterpart RetinaNet by 2.4% in AP with the same backbone. FCOS also outperforms the recent anchor-free one-stage detector CornerNet with much less design complexity. Refer to Table 3 for details of “improvements”.

| Method                     | # samples | AR <sup>100</sup> | AR <sup>1k</sup> |
|----------------------------|-----------|-------------------|------------------|
| RPN w/ FPN & GN (ReImpl.)  | ~200K     | 44.7              | 56.9             |
| FCOS w/ GN w/o center-ness | ~66K      | 48.0              | 59.3             |
| FCOS w/ GN                 | ~66K      | <b>52.8</b>       | <b>60.3</b>      |

**Table 6** – FCOS as Region Proposal Networks vs. RPNs with FPN. ResNet-50 is used as the backbone. FCOS improves AR<sup>100</sup> and AR<sup>1k</sup> by 8.1% and 3.4%, respectively. GN: Group Normalization.

(GN) [29] in the newly added convolutional layers except for the last prediction layers, which makes our training more stable. 2) We use  $P_5$  to produce the  $P_6$  and  $P_7$  instead of  $C_5$  in the standard RetinaNet. We observe that using  $P_5$  can improve the performance slightly.

To show that our FCOS can serve as a simple and strong alternative of anchor-based detectors, and for a fair comparison, we remove GN (the gradients are clipped to prevent them from exploding) and use  $C_5$  in our detector. As shown in Table 3, with exactly the same settings, our FCOS still compares favorably with the anchor-based detector (36.3% vs 35.9%). Moreover, it is worth to note that we directly use all hyper-parameters (*e.g.*, learning rate, the NMS threshold and etc.) from RetinaNet, which have been optimized for the anchor-based detector. We argue that the performance of FCOS can be improved further if the hyper-parameters are tuned for it.

It is worth noting that with some almost cost-free improvements, as shown in Table 3, the performance of our anchor-free detector can be improved by a large margin. Given the superior performance and the merits of the anchor-free detector (*e.g.*, much simpler and fewer hyper-parameters than anchor-based detectors), we encourage the community to rethink the necessity of anchor boxes in object detection.

## 4.2. Comparison with State-of-the-art Detectors

We compare FCOS with other state-of-the-art object detectors on *test* – *dev* split of MS-COCO benchmark. For these experiments, we randomly scale the shorter side of images in the range from 640 to 800 during the training and double the number of iterations to 180K (with the learning rate change points scaled proportionally). Other settings are exactly the same as the model with AP 37.1% in Table 3. As shown in Table 5, with ResNet-101-FPN, our FCOS outperforms the RetinaNet with the same backbone ResNet-101-FPN by 2.4% in AP. To our knowledge, it is the first time that an anchor-free detector, without any bells and whistles outperforms anchor-based detectors by a large margin. FCOS also outperforms other classical two-stage anchor-based detectors such as Faster R-CNN by a large margin. With ResNeXt-64x4d-101-FPN [30] as the backbone, FCOS achieves 43.2% in AP. It outperforms the recent state-of-the-art anchor-free detector CornerNet [13] by a large margin while being much simpler. Note that CornerNet requires to group corners with embedding vectors, which needs special design for the detector. Thus, we argue that FCOS is more likely to serve as a strong and simple alternative to current mainstream anchor-based detectors. Moreover, FCOS with the improvements in Table 3 achieves 44.7% in AP with single-model and single scale testing, which surpasses previous detectors by a large margin.

## 5. Extensions on Region Proposal Networks

So far we have shown that in a one-stage detector, our FCOS can achieve even better performance than anchor-based counterparts. Intuitively, FCOS should be also able

| Method                     | Backbone                 | AP          | AP <sub>50</sub> | AP <sub>75</sub> | AP <sub>S</sub> | AP <sub>M</sub> | AP <sub>L</sub> |
|----------------------------|--------------------------|-------------|------------------|------------------|-----------------|-----------------|-----------------|
| Two-stage methods:         |                          |             |                  |                  |                 |                 |                 |
| Faster R-CNN w/ FPN [14]   | ResNet-101-FPN           | 36.2        | 59.1             | 39.0             | 18.2            | 39.0            | 48.2            |
| Faster R-CNN by G-RMI [11] | Inception-ResNet-v2 [27] | 34.7        | 55.5             | 36.7             | 13.5            | 38.1            | 52.0            |
| Faster R-CNN w/ TDM [25]   | Inception-ResNet-v2-TDM  | 36.8        | 57.7             | 39.2             | 16.2            | 39.8            | 52.1            |
| One-stage methods:         |                          |             |                  |                  |                 |                 |                 |
| YOLOv2 [22]                | DarkNet-19 [22]          | 21.6        | 44.0             | 19.2             | 5.0             | 22.4            | 35.5            |
| SSD513 [18]                | ResNet-101-SSD           | 31.2        | 50.4             | 33.3             | 10.2            | 34.5            | 49.8            |
| DSSD513 [5]                | ResNet-101-DSSD          | 33.2        | 53.3             | 35.2             | 13.0            | 35.4            | 51.1            |
| RetinaNet [15]             | ResNet-101-FPN           | 39.1        | 59.1             | 42.3             | 21.8            | 42.7            | 50.2            |
| CornerNet [13]             | Hourglass-104            | 40.5        | 56.5             | 43.1             | 19.4            | 42.7            | 53.9            |
| FSAF [34]                  | ResNeXt-64x4d-101-FPN    | 42.9        | 63.8             | 46.3             | 26.6            | 46.2            | 52.7            |
| FCOS                       | ResNet-101-FPN           | 41.5        | 60.7             | 45.0             | 24.4            | 44.8            | 51.6            |
| FCOS                       | HRNet-W32-51 [26]        | 42.0        | 60.4             | 45.3             | 25.4            | 45.0            | 51.0            |
| FCOS                       | ResNeXt-32x8d-101-FPN    | 42.7        | 62.2             | 46.1             | 26.0            | 45.6            | 52.6            |
| FCOS                       | ResNeXt-64x4d-101-FPN    | 43.2        | 62.8             | 46.6             | 26.5            | 46.2            | 53.3            |
| FCOS w/ improvements       | ResNeXt-64x4d-101-FPN    | <b>44.7</b> | <b>64.1</b>      | <b>48.4</b>      | <b>27.6</b>     | <b>47.5</b>     | <b>55.6</b>     |

表5 – FCOS与其他先进的两阶段或一阶段检测器对比 (*single-model and single-scale results*)。在相同骨干网络下, FCOS的AP比基于锚点的对应方法RetinaNet高出2.4%。FCOS也优于近期无需锚点的单阶段检测器CornerNet, 且设计复杂度显著降低。具体“改进”细节请参见表3。

| Method                     | # samples | AR <sup>100</sup> | AR <sup>1k</sup> |
|----------------------------|-----------|-------------------|------------------|
| RPN w/ FPN & GN (ReImpl.)  | ~200K     | 44.7              | 56.9             |
| FCOS w/ GN w/o center-ness | ~66K      | 48.0              | 59.3             |
| FCOS w/ GN                 | ~66K      | <b>52.8</b>       | <b>60.3</b>      |

表6 – FCOS作为区域提议网络与使用FPN的RPN对比。主干网络采用ResNet-50。FCOS将AR<sup>100</sup>和AR<sup>1k</sup>分别提升了8.1%和3.4%。GN: 组归一化。

(GN) [29] 在新增加的卷积层中 (除了最后的预测层), 这使得我们的训练更加稳定。2) 我们使用  $P_5$  来生成  $P_6$  和  $P_7$ , 而不是标准 RetinaNet 中的  $C_5$ 。我们观察到使用  $P_5$  可以略微提升性能。

为了证明我们的FCOS可以作为基于锚点检测器的一个简单而强大的替代方案, 并且为了公平比较, 我们移除了GN (梯度被裁剪以防止爆炸) 并在检测器中使用了  $C_5$ 。如表3所示, 在完全相同的设置下, 我们的FCOS仍然优于基于锚点的检测器 (36.3% vs 35.9%)。此外, 值得注意的是, 我们直接使用了RetinaNet的所有超参数 (*e.g.*、学习率、NMS阈值等), 这些参数已针对基于锚点的检测器进行了优化。我们认为, 如果为FCOS调整超参数, 其性能还可以进一步提升。

值得注意的是, 通过一些几乎零成本的改进 (如表3所示), 我们的无锚点检测器性能可获得大幅提升。鉴于无锚点检测器的卓越性能及其优势 ( $\{v^*\}$ , 比基于锚点的检测器更简洁且超参数更少), 我们建议学界重新思考目标检测中锚框的必要性。

## 4.2. 与最先进检测器的比较

我们在MS-COCO基准测试的test – dev分割上, 将FCOS与其他最先进的物体检测器进行比较。在这些实验中, 我们在训练期间将图像的短边随机缩放至640到800的范围内, 并将迭代次数加倍至180K (学习率变化点按比例调整)。其他设置与表3中AP为37.1%的模型完全相同。如表5所示, 使用ResNet-101-FPN作为骨干网络时, 我们的FCOS在AP上比使用相同骨干网络ResNet-101-FPN的RetinaNet高出2.4%。据我们所知, 这是首次有无锚框检测器在没有任何额外修饰的情况下, 以较大优势超越基于锚框的检测器。FCOS同样以较大优势超越了其他经典的基于锚框的两阶段检测器, 如Faster R-CNN。使用ResNeXt-64x4d-101-FPN[30]作为骨干网络时, FCOS实现了43.2%的AP。它以较大优势超越了近期最先进的无锚框检测器CornerNet[13], 同时结构更为简洁。值得注意的是, CornerNet需要通过嵌入向量对角点进行分组, 这需要为检测器进行特殊设计。因此, 我们认为FCOS更有可能成为当前主流基于锚框检测器的强大而简洁的替代方案。此外, 结合表3中的改进, FCOS在单模型和单尺度测试下实现了44.7%的AP, 以较大优势超越了先前的检测器。

## 5. 区域提议网络的扩展

到目前为止, 我们已经证明, 在单阶段检测器中, 我们的FCOS可以实现比基于锚点的同类方法更好的性能。直观上, FCOS应该也能够

to replace the anchor boxes in Region Proposal Networks (RPNs) with FPN [14] in the two-stage detector Faster R-CNN. Here, we confirm that by experiments.

Compared to RPNs with FPN [14], we replace anchor boxes with the method in FCOS. Moreover, we add GN into the layers in FPN heads, which can make our training more stable. All other settings are exactly the same with RPNs with FPN in the official code [7]. As shown in Table 6, even without the proposed center-ness branch, our FCOS already improves both  $AR^{100}$  and  $AR^{1k}$  significantly. With the proposed center-ness branch, FCOS further boosts  $AR^{100}$  and  $AR^{1k}$  respectively to 52.8% and 60.3%, which are 18% relative improvement for  $AR^{100}$  and 3.4% absolute improvement for  $AR^{1k}$  over the RPNs with FPN.

## 6. Conclusion

We have proposed an anchor-free and proposal-free one-stage detector FCOS. As shown in experiments, FCOS compares favourably against the popular anchor-based one-stage detectors, including RetinaNet, YOLO and SSD, but with much less design complexity. FCOS completely avoids all computation and hyper-parameters related to anchor boxes and solves the object detection in a per-pixel prediction fashion, similar to other dense prediction tasks such as semantic segmentation. FCOS also achieves state-of-the-art performance among one-stage detectors. We also show that FCOS can be used as RPNs in the two-stage detector Faster R-CNN and outperforms the its RPNs by a large margin. Given its effectiveness and efficiency, we hope that FCOS can serve as a strong and simple alternative of current mainstream anchor-based detectors. We also believe that FCOS can be extended to solve many other instance-level recognition tasks.

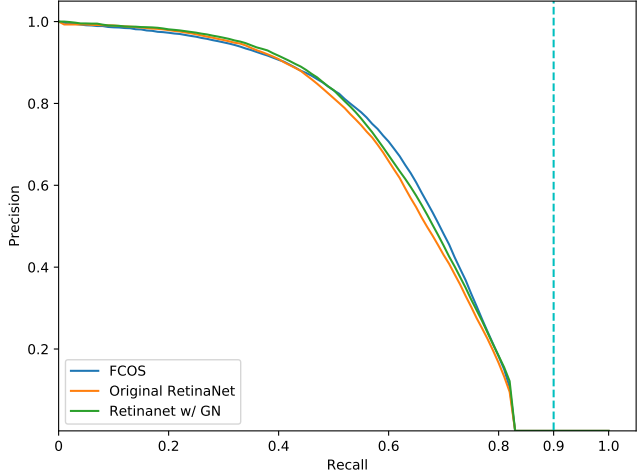
## Appendix

### 7. Class-agnostic Precision-recall Curves

| Method                 | AP          | AP <sub>50</sub> | AP <sub>75</sub> | AP <sub>90</sub> |
|------------------------|-------------|------------------|------------------|------------------|
| Orginal RetinaNet [15] | 39.5        | 63.6             | 41.8             | 10.6             |
| RetinaNet w/ GN [29]   | 40.0        | 64.5             | 42.2             | 10.4             |
| FCOS                   | <b>40.5</b> | <b>64.7</b>      | <b>42.6</b>      | <b>13.1</b>      |
|                        |             | +0.2             | +0.4             | +2.7             |

**Table 7** – The class-agnostic detection performance for RetinaNet and FCOS. FCOS has better performance than RetinaNet. Moreover, the improvement over RetinaNet becomes larger with a stricter IOU threshold. The results are obtained with the same models in Table 4 of our main paper.

In Fig. 4, Fig. 5 and Fig. 6, we present class-agnostic precision-recall curves on split minival at IOU thresholds



**Figure 4** – Class-agnostic precision-recall curves at IOU = 0.50.

being 0.50, 0.75 and 0.90, respectively. Table 7 shows APs corresponding to the three curves.

As shown in Table 7, our FCOS achieves better performance than its anchor-based counterpart RetinaNet. Moreover, it worth noting that with a stricter IOU threshold, FCOS enjoys a larger improvement over RetinaNet, which suggests that FCOS has a better bounding box regressor to detect objects more accurately. One of the reasons should be that FCOS has the ability to leverage more foreground samples to train the regressor as mentioned in our main paper.

Finally, as shown in all precision-recall curves, the best recalls of these detectors in the precision-recall curves are much lower than 90%. It further suggests that the small gap (98.40% vs. 99.23%) of *best possible recall (BPR)* between FCOS and RetinaNet hardly harms the final detection performance.

### 8. Visualization for Center-ness

As mentioned in our main paper, by suppressing low-quality detected bounding boxes, the proposed center-ness branch improves the detection performance by a large margin. In this section, we confirm this.

We expect that the center-ness can down-weight the scores of low-quality bounding boxes such that these bounding boxes can be filtered out in following post-processing such as non-maximum suppression (NMS). A detected bounding box is considered as a low-quality one if it has a low IOU score with its corresponding ground-truth bounding box. A bounding box with low IOU but a high confidence score is likely to become a false positive and harm the precision.

In Fig. 7, we consider a detected bounding box as a 2D

在两级检测器Faster R-CNN中，用FPN [14] 替换区域提议网络（RPNs）中的锚框。我们通过实验对此进行了验证。

与采用FPN的RPN[14]相比，我们使用FCOS中的方法替换了锚框。此外，我们在FPN头部各层中加入了GN，这能使训练更加稳定。其余所有设置均与官方代码[7]中采用FPN的RPN完全相同。如表6所示，即使未使用提出的中心度分支，我们的FCOS也已显著提升了 $AR^{100}$ 和 $AR^{1k}$ 。加入提出的中心度分支后，FCOS进一步将 $AR^{100}$ 和 $AR^{1k}$ 分别提升至52.8%和60.3%，相较于采用FPN的RPN， $AR^{100}$ 实现了18%的相对提升， $AR^{1k}$ 则获得3.4%的绝对提升。

## 6. 结论

我们提出了一种无锚框且无需候选区域的单阶段检测器FCOS。实验表明，FCOS与当前流行的基于锚框的单阶段检测器（包括RetinaNet、YOLO和SSD）相比具有优势，同时设计复杂度显著降低。FCOS完全避免了与锚框相关的所有计算和超参数，并以逐像素预测的方式解决目标检测问题，其形式类似于语义分割等其他密集预测任务。FCOS在单阶段检测器中实现了最先进的性能。我们还证明FCOS可作为两阶段检测器Faster R-CNN中的区域提议网络（RPN），并以显著优势超越其原有RPN。鉴于其高效性与有效性，我们希望FCOS能成为当前主流基于锚框检测器的有力且简洁的替代方案。我们亦相信FCOS可扩展应用于解决其他实例级识别任务。

## 附录

### 7. 类别无关的精确率-召回率曲线

| Method                 | AP          | AP <sub>50</sub> | AP <sub>75</sub> | AP <sub>90</sub> |
|------------------------|-------------|------------------|------------------|------------------|
| Orginal RetinaNet [15] | 39.5        | 63.6             | 41.8             | 10.6             |
| RetinaNet w/ GN [29]   | 40.0        | 64.5             | 42.2             | 10.4             |
| FCOS                   | <b>40.5</b> | <b>64.7</b>      | <b>42.6</b>      | <b>13.1</b>      |
|                        |             | +0.2             | +0.4             | +2.7             |

表7 – RetinaNet和FCOS的类别无关检测性能。FCOS的性能优于RetinaNet。此外，随着IOU阈值更严格，FCOS相对于RetinaNet的提升幅度更大。这些结果采用我们主论文表4中的相同模型获得。

在图4、图5和图6中，我们展示了在IOU阈值为 $\{v^*\}$ 时，针对分割minival的类别无关精确率-召回率曲线。

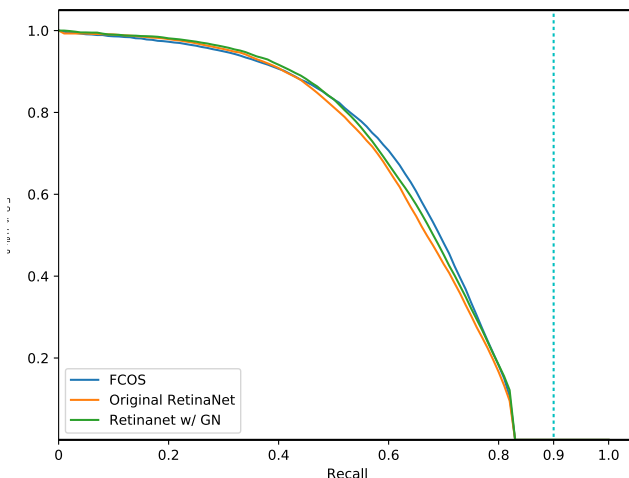


图4 – 在IOU  $\{v^*\}$  0.50下的类别无关精确率-召回率曲线。

分别为0.50、0.75和0.90。表7展示了这三条曲线对应的AP值。

如表7所示，我们的FCOS相比其基于锚点的对应方法RetinaNet取得了更优的性能。值得注意的是，在更严格的IOU阈值下，FCOS相对于RetinaNet的提升幅度更大，这表明FCOS拥有更优的边界框回归器，能够更精确地检测目标。原因之一应是FCOS能够如主论文所述，利用更多前景样本来训练回归器。

最后，如所有精确率-召回率曲线所示，这些检测器在曲线中的最佳召回率远低于90%。这进一步表明，FCOS与RetinaNet之间*best possible recall (BPR)*的微小差距（98.40%对比99.23%）几乎不会损害最终检测性能。

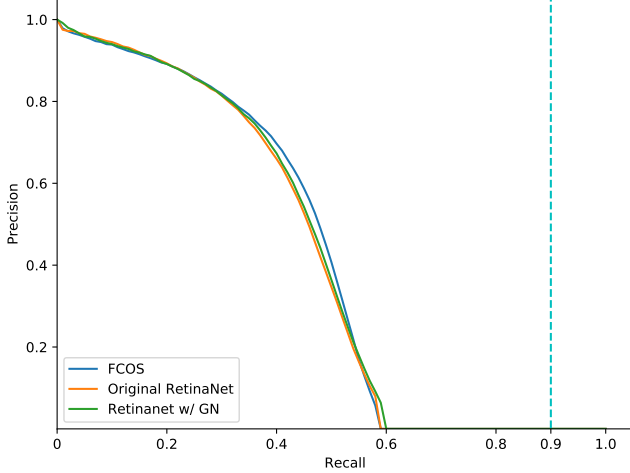
### 8. 中心度可视化

如我们在主论文中所述，通过抑制低质量检测边界框，所提出的中心度分支大幅提升了检测性能。本节将对此进行验证。

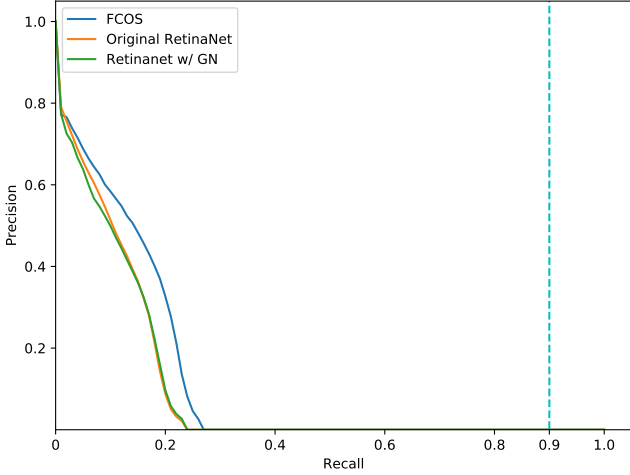
我们期望中心度能够降低低质量边界框的得分，使得这些边界框在后续的后处理（如非极大值抑制）中被过滤掉。如果一个检测到的边界框与其对应的真实边界框的交并比得分较低，那么它就被认为是低质量的。一个交并比低但置信度得分高的边界框很可能成为误报，从而损害精确度。

在图7中，我们将检测到的边界框视为一个二维





**Figure 5** – Class-agnostic precision-recall curves at IOU = 0.75.



**Figure 6** – Class-agnostic precision-recall curves at IOU = 0.90.

point  $(x, y)$  with  $x$  being its score and  $y$  being the IOU with its corresponding ground-truth box. As shown in Fig. 7 (left), before applying the center-ness, there are a large number of low-quality bounding boxes but with a high confidence score (i.e., the points under the line  $y = x$ ). Due to their high scores, these low-quality bounding boxes cannot be eliminated in post-processing and result in lowering the precision of the detector. After multiplying the classification score with the center-ness score, these points are pushed to the left side of the plot (i.e., their scores are reduced), as shown in Fig. 7 (right). As a result, these low-quality bounding boxes are much more likely to be filtered out in post-processing and the final detection performance can be improved.

## 9. Qualitative Results

Some qualitative results are shown in Fig. 8. As shown in the figure, our proposed FCOS can detect a wide range of objects including crowded, occluded, highly overlapped, extremely small and very large objects.

## 10. More discussions

### Center-ness vs. IoUNet:

Center-ness and IoUNet of Jiang et al. “Acquisition of Localization Confidence for Accurate Object Detection” shares a similar purpose (i.e., to suppress low-quality predictions) with different approaches. IoUNet trains a separate network to predict the IoU score between predicted bounding-boxes and ground-truth boxes. Center-ness, as a part of our detector, only has a single layer and is trained jointly with the detector, thus being much simpler. Moreover, “center-ness” does not take as input the predicted bounding-boxes. Instead, it directly accesses the location’s ability to predict high-quality bounding-boxes.

### BPR in Section 4.1 and ambiguity analysis:

We do not aim to compare “recall by specific IoU” with “recall by pixel within box”. The main purpose of Table 1 is to show that the upper bound of recall of FCOS is very close to the upper bound of recall of anchor-based RetinaNet (98.4% vs. 99.23%). BPR by other IoU thresholds are listed as those are used in the official code of RetinaNet. Moreover, *no evidence shows that the regression targets of FCOS are difficult to learn because they are more spread-out*. FCOS in fact yields more accurate bounding-boxes.

During training, we deal with the ambiguity at the same FPN level by choosing the ground-truth box with the minimal area. When testing, if two objects A and B with the same class have overlap, no matter which one object the locations in the overlap predict, the prediction is correct and the missed one can be predicted by the locations only belonging to it. In the case that A and B do not belong to the same class, a location in the overlap might predict A’s class but regress B’s bounding-box, which is a mistake. That is why we only count the ambiguity across different classes. Moreover, it appears that this ambiguity does not make FCOS worse than RetinaNet in AP, as shown in Table 8.

### Additional ablation study:

As shown in Table 8, a vanilla FCOS performs on par with RetinaNet, being of simpler design and with  $\sim 9\times$  less network outputs. Moreover, FCOS works much better than RetinaNet with single anchor. As for the 2% gain on test-dev, besides the performance gain brought by the components in Table 8, we conjecture that different training details (e.g., learning rate schedule) might cause slight differences in performance.

### RetinaNet with Center-ness:

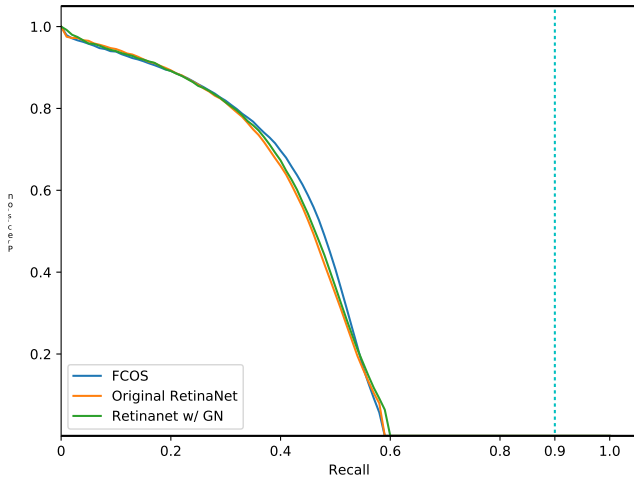


图5 – 在IOU = 0.75下的类别无关精确率-召回率曲线。

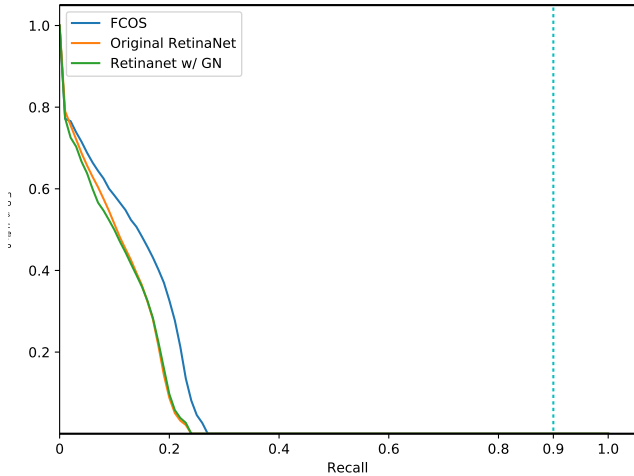


图6 – 在IOU = 0.90下的类别无关精确率-召回率曲线。

点 $(x, y)$ ，其中 $x$ 为其得分， $y$ 为其与对应真实框的交并比。如图7（左）所示，在应用中心度之前，存在大量低质量边界框却具有高置信度得分（即位于直线 $y = x$ 下方的点）。由于得分较高，这些低质量边界框无法在后处理中被消除，导致检测器的精度下降。将分类得分与中心度得分相乘后，这些点被推至图的左侧（即其得分降低），如图7（右）所示。因此，这些低质量边界框在后处理中被过滤掉的可能性大大增加，最终检测性能得以提升。

## 9. 定性结果

一些定性结果如图8所示。如图所示，我们提出的FCOS能够检测包括拥挤、遮挡、高度重叠、极小和极大物体在内的广泛目标。

## 10. 更多讨论

中心度 vs. IoUNet:

Jiang等人的“Acquisition of Localization Confidence for Accurate Object Detection”中的Center-ness和IoUNet具有相似的目的（*i.e.*，即抑制低质量预测），但方法不同。IoUNet训练一个独立的网络来预测预测边界框与真实边界框之间的IoU分数。而Center-ness作为我们检测器的一部分，仅包含单层结构并与检测器联合训练，因此更为简洁。此外，“center-ness”并不以预测边界框作为输入，而是直接评估位置预测高质量边界框的能力。

第4.1节中的BPR与歧义分析:

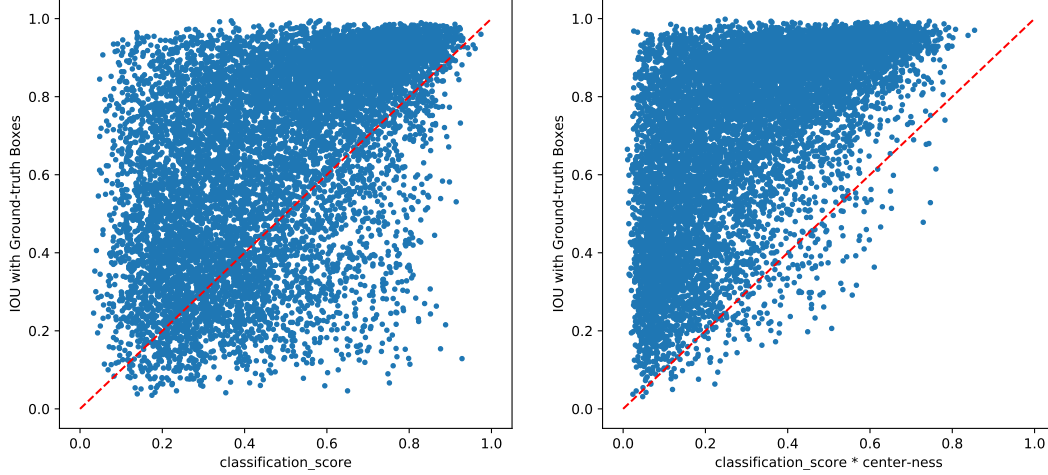
我们并不旨在比较“特定IoU下的召回率”与“框内像素召回率”。表1的主要目的是展示FCOS的召回率上限与基于锚点的RetinaNet的召回率上限非常接近（98.4%对比99.23%）。其他IoU阈值下的BPR也被列出，因为这些阈值在RetinaNet官方代码中被使用。此外，*no evidence shows that the regression targets of FCOS are difficult to learn because they are more spread-out*. FCOS实际上能产生更准确的边界框。

在训练过程中，我们通过选择面积最小的真实框来处理同一FPN层级上的歧义。测试时，若两个同类物体A和B存在重叠区域，则无论重叠区域内的位置点预测哪一个物体，该预测均被视为正确，而未被预测到的物体仍可通过其专属位置点进行预测。当A和B属于不同类别时，重叠区域内的位置点可能出现预测为A类别却回归到B边界框的错误情况。因此我们仅统计跨类别的歧义现象。此外，如表8所示，这种歧义并未导致FCOS在AP指标上逊色于RetinaNet。

额外的消融研究:

如表8所示，基础版FCOS与RetinaNet表现相当，其设计更简洁且网络输出减少 $\sim 9\times$ 。此外，FCOS在单锚点设置下明显优于RetinaNet。关于test-dev上2%的性能提升，除表8中各组件带来的增益外，我们推测不同的训练细节（*e.g.*、学习率调整策略）可能导致性能的细微差异。

带中心度的RetinaNet:



**Figure 7** – Without (left) or with (right) the proposed center-ness. A point in the figure denotes a detected bounding box. The dashed line is the line  $y = x$ . As shown in the figure (right), after multiplying the classification scores with the center-ness scores, the low-quality boxes (under the line  $y = x$ ) are pushed to the left side of the plot. It suggests that the scores of these boxes are reduced substantially.

| Method           | $C_5/P_5$ | GN | Scalar | IoU | AP          |
|------------------|-----------|----|--------|-----|-------------|
| RetinaNet (#A=1) | $C_5$     |    |        |     | 32.5        |
| RetinaNet (#A=9) | $C_5$     |    |        |     | 35.7        |
| FCOS (pure)      | $C_5$     |    |        |     | 35.7        |
| FCOS             | $P_5$     |    |        |     | 35.8        |
| FCOS             | $P_5$     | ✓  |        |     | 36.3        |
| FCOS             | $P_5$     | ✓  | ✓      |     | 36.4        |
| FCOS             | $P_5$     | ✓  | ✓      | ✓   | <b>36.6</b> |

**Table 8** – Ablation study on MS-COCO minival. “#A” is the number of anchor boxes per location in RetinaNet. “IOU” is IOU loss. “Scalar” denotes whether to use scalars in exp. All experiments are conducted with the same settings.

Center-ness cannot be directly used in RetinaNet with multiple anchor boxes per location because one location on feature maps has *only one* center-ness score but different anchor boxes on the same location require different “center-ness” (note that center-ness is also used as “soft” thresholds for positive/negative samples).

For anchor-based RetinaNet, the IoU score between anchor boxes and ground-truth boxes may serve as an alternative of “center-ness”.

#### Positive samples overlap with RetinaNet:

We want to highlight that center-ness comes into play *only* when testing. When training, all locations within ground-truth boxes are marked as positive samples. As a result, FCOS can use more foreground locations to train the regressor and thus yield more accurate bounding-boxes.

**Acknowledgments.** We would like to thank the author of [1] for the tricks of center sampling and GIoU. We also

thank Chaorui Deng for HRNet based FCOS and his suggestion of positioning the center-ness branch with box regression.

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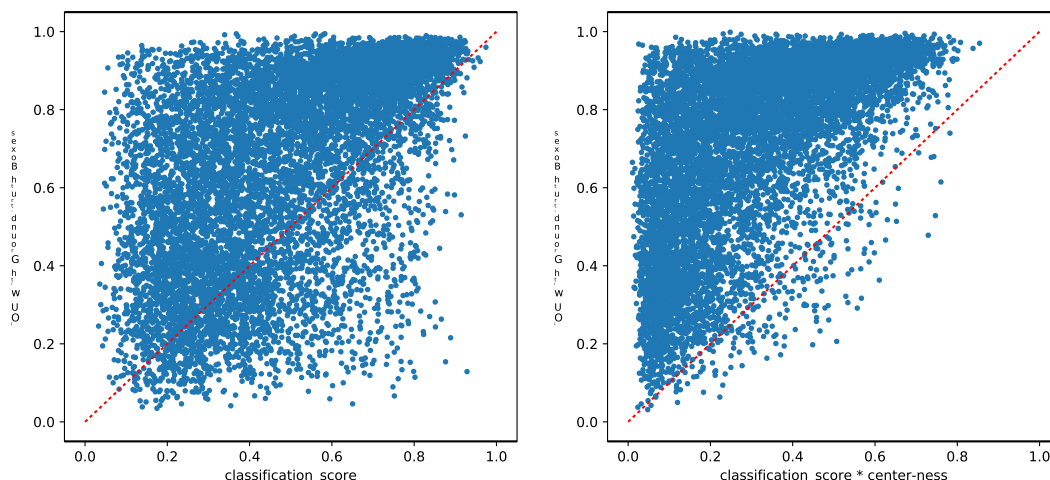


图7 – 无（左）或有（右）所提出的中心度。图中的点表示检测到的边界框。虚线为直线  $y = x$ 。如图所示（右），将分类分数与中心度分数相乘后，低质量框（直线  $y = x$  下方）被推至图的左侧。这表明这些框的分数被显著降低。

| Method           | $C_5/P_5$ | GN | Scalar | IoU | AP          |
|------------------|-----------|----|--------|-----|-------------|
| RetinaNet (#A=1) | $C_5$     |    |        |     | 32.5        |
| RetinaNet (#A=9) | $C_5$     |    |        |     | 35.7        |
| FCOS (pure)      | $C_5$     |    |        |     | 35.7        |
| FCOS             | $P_5$     |    |        |     | 35.8        |
| FCOS             | $P_5$     | ✓  |        |     | 36.3        |
| FCOS             | $P_5$     | ✓  | ✓      |     | 36.4        |
| FCOS             | $P_5$     | ✓  | ✓      | ✓   | <b>36.6</b> |

表8 – 在MS-COCO数据集上的消融研究 *minival*。“#A”表示RetinaNet中每个位置的锚框数量。“IOU”指IOU损失函数。“Scalar”表示是否在指数运算中使用标量。所有实验均在相同设置下进行。

中心度无法直接用于每个位置有多个锚框的RetinaNet中，因为特征图上的一个位置只有一个 *only one* 中心度分数，但同一位置的不同锚框需要不同的“中心度”（请注意，中心度也用作正负样本的“软”阈值）。

对于基于锚点的RetinaNet，锚框与真实框之间的IoU分数可以作为“中心度”的一种替代指标。

正样本与RetinaNet重叠：

我们想强调的是，中心度在测试时发挥作用 *only*。在训练时，所有位于真实框内的位置都被标记为正样本。因此，FCOS可以利用更多的前景位置来训练回归器，从而产生更准确的边界框。

致谢。我们要感谢[1]的作者提出的中心采样和GIoU技巧。我们同样

感谢邓超瑞基于HRNet的FCOS以及他将中心度分支与边界框回归结合的建议。

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**Figure 8** – Some detection results on *minival* split. ResNet-50 is used as the backbone. As shown in the figure, FCOS works well with a wide range of objects including crowded, occluded, highly overlapped, extremely small and very large objects.

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图 8 – 在 *minival* 分割集上的部分检测结果。采用 ResNet-50 作为骨干网络。如图所示，FCOS 在包括密集、遮挡、高度重叠、极小和极大物体在内的多种目标上均表现良好。

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